Land-Use Litigation and the Microgeography of NIMBYism

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As cities grow, building in already populated areas becomes central to housing affordability and land preservation, but also triggers not-in-my-backyard (NIMBY) opposition. To measure it, we construct a dataset from judicial archives covering all construction and land-use litigation in France since 2005. Using an LLM, we extract with high accuracy rich contextual information from over 200,000 court decisions. We document that NIMBY opposition primarily takes the form of individuals challenging housing permits, disproportionately targeting multifamily housing. Consistent with a simple model in which neighbors litigate to protect their property values, appeals to approved housing concentrate in dense and expensive areas. Second-home owners also account for a significant share of litigation, especially in high-end tourist locations. In low-density areas, by contrast, litigation more often comes from owners of undeveloped land appealing permit denials to preserve their building rights.

Keywords: Land-use conflicts; NIMBY; housing supply; LLM; litigation

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1 INTRODUCTION

Urban growth involves a trade-off between housing affordability, now a major concern in many economies (UN Habitat, 2026), and urban sprawl, a well-documented source of environmental degradation (IPBES, 2018). A common policy response is to promote denser development within already built-up areas, so that new housing supply can keep pace with demand while preserving natural land. Such initiatives, however, often face local opposition. Since new housing has diffuse benefits and localized costs, residents tend to prefer construction taking place away from their own homes, and this intensifies when new supply comes at high density. This local opposition, commonly referred to as NIMBY ("Not In My Backyard"), has been shown to act as a "regulatory tax" on new housing (Glaeser et al., 2005a).

In this paper, we focus on a particularly direct form of NIMBY behavior that bypasses the local electoral process: individuals challenging new construction in court. Litigation matters because it requires neither a majority nor an organized group, yet individual appeals add up to mass opposition (Melot, 2015). Moreover, local opposition depends on litigants' resources and willingness to appeal rather than on collective preferences. Thus, even local majorities may face large implementation costs when adopting pro-housing policies.

We introduce a new dataset covering all litigation over land-use authorizations in France between 2005 and 2023. Using a Large Language Model (LLM), we extract 20 variables from the text of 200,000 administrative court decisions, covering the legal claims and the litigants, the location and characteristics of the contested construction, and the outcome of the ruling. The LLM consistently achieved high performance, with accuracy rates exceeding 95% for more than half of the variables, and above 83% for all. France is particularly well suited to this exercise because the archives of its forty-two administrative courts are centralized, which allows us to cover litigation across the whole country. Moreover, because the municipality that granted or denied the permit is involved in the court case, we can link each case to one of the 35,000 French municipalities. We thus have information on litigation at a fine geographic level with detailed census data.

The majority of litigation cases are filed by third-party individuals against new housing, rather than large infrastructure projects commonly associated with environmental harm. These appeals, which we label *Anti-construction*, disproportionally target denser forms of housing: multifamily housing accounts for 83% of contested units but only 51% of approved units over the period. These cases take 16 months on average to be decided and result in annulment 32% of the time, enough to constitute a credible threat to new projects. We also observe a substantial number of *Pro-construction* appeals filed against permit denials. On average they are relatively more frequent than *Anti-construction* litigation, which we interpret as reflecting the "own-lot effect", the important loss associated with reduced building rights (Turner et al., 2014). These cases also provide a benchmark: if litigation simply reflected a general tendency to challenge local decisions, both types should exhibit similar patterns.

To investigate the determinants of these appeals, we develop a simple framework in which neighbors appeal to protect their home values from a new project whenever the expected gains from delay, annulment, or a settlement with the developer exceed litigation costs. Because both the neighbors' potential loss and the developer's stake rise with local property values and project size, the framework predicts that *Anti-construction* litigation increases with prices and project scale, as well as with the density of neighbors, which raises the number of potential litigants. *Pro-construction* litigation, by contrast, depends solely on project's own value.

In our empirical analysis, we measure the intensity of litigation as the share of permits appealed. On average, 1.3% (resp. 1.9%) of all approved (resp. denied) permits and 7.6% (resp. 4.5%) of all approved (resp. denied) units face an appeal. There is substantial spatial heterogeneity, with many areas without any litigation while 12% of approved units are appealed in the Paris metropolitan area and as many as 39%

in Saint-Tropez. These figures are comparable to California, where environmental-review lawsuits target at least 7% of residential units (Hernandez, 2022) and 36% of units in Los Angeles approved between 2014 and 2016 face administrative appeals (Biber et al., 2019). We then regress local litigation rates on municipal characteristics to identify which factors drive this concentration. Density, housing prices and multifamily construction are the main determinants of *Anti-construction* litigation, jointly accounting for two-thirds of the within-urban-area variation. Homeownership, by contrast, is not significant. Multiple property owners also play a significant role: the coefficient on the share of secondary homes is positive and especially large in smaller urban areas. Using litigants' addresses, available until 2013, we show that a large part of the secondary-homes effect comes from their owners themselves: removing the litigants who do not live in the municipality significantly weakens the corresponding coefficient. Second-home owners may therefore not only raise prices on the demand side (Hilber & Schöni, 2020) but also restrict supply through litigation. Taken together, these results echo the "influential landowner" hypothesis of Hilber and Robert-Nicoud (2013): owners of developed land litigate against construction, while owners of undeveloped land litigate for their building rights. Because *Anti-construction* concentrates where housing demand is strongest, it further pushes new development toward the urban fringe.

Our paper most directly relates to the literature on the political economy of housing supply. Following the *homevoter hypothesis* of Fischel (2001), several studies model restrictive regulation as the outcome of homeowners' influence over local governments through the electoral process (Feinerman et al., 2004; Ortalo-Magné & Prat, 2014) or via competition between interest groups of landowners (Glaeser et al., 2005b; Hilber & Robert-Nicoud, 2013). A fruitful empirical literature has established a causal link between local political incentives and housing supply policies[4]. We extend this literature by demonstrating that the political channel does not capture the full extent of NIMBYism. Recent studies of construction delays point to constraints that operate outside local decision-making without measuring them directly (D'Amico et al., 2026; Lyons & Sweeney, 2025), one notable exception being Gandhi et al. (2021) who measure the effect of litigation on large condominium projects in Mumbai. We provide nationwide, systematic evidence on this channel in a developed country setting: we measure how much opposition persists once projects have been approved and identify its local determinants.

The dataset we introduce also contributes to the literature on the measure of local constraints on construction. While recent studies in economics have focused on zoning and land-use regulation (Bartik et al., 2023; Büchler & v. Ehrlich, 2023; Gyourko et al., 2008; Song, 2025), a broader literature studies NIMBYism and attitudes toward different types of construction through surveys and experiments (Larsen & Nyholt, 2026; Monkkonen & Manville, 2019; Trounstine, 2023). Our work is closer to the latter but captures preferences revealed through costly appeals against building permits. This work adds to a rapidly growing literature in economics that uses LLMs to turn text into data for quantitative analysis (Dell, 2025). It is noteworthy that judicial decisions are structured enough for extraction to achieve very high accuracy, opening up forms of opposition that have been difficult to capture empirically.

The rest of the article is organized as follows. Section 2 presents the institutional context and methodology used to build the litigation database. Section 3 provides an anatomy of land-use litigation. Section 4 proposes a theoretical model of appeal. Section 5 examines the economic determinants of NIMBYism. Section 6 offers concluding remarks.

4 Kahn (2011) shows that more politically liberal cities in California permit less new housing. Solé-Ollé and Viladecans-Marsal (2012) find that stronger political competition reduces the amount of land Spanish municipalities designate for development. Mast (2024) finds that US towns switching from at-large to ward-based council elections permit about 20% fewer housing units. Fang et al. (2023) show that municipal councillors in Toronto who represent homeowners are less likely to approve housing bills. Tricaud (2025) shows that French municipalities forced into intermunicipal communities saw a 12.5% increase in building permits.

2 INSTITUTIONAL CONTEXT AND DATA

This section describes the institutional context of land-use authorizations in France and the construction of the litigation database as well as other key datasets we rely on.

2.1 Institutional context

Since the decentralization reforms of the early 1980s, French municipalities have been responsible for urban planning within their boundaries. Each municipality drafts and adopts a planning document, and the mayor then grants building, demolition and subdivision permits in strict conformity with that plan. Citizens who disagree with a municipal decision can try to negotiate with the municipality first, but they may also take the matter directly to court.

Litigation concerning the legality of land-use permissions is governed by the section of Administrative Law embodied in the *Code de l'Urbanisme*. It is handled by the forty-two administrative courts, which are located in the main cities of the different regions. The mass of land-use litigation has been increasing steadily and accounts for about 5-7% of the caseload of administrative courts (Conseil d'État, 2024).

Judicial appeals are known to have adverse effects on construction activity. In France in particular, notaries generally do not sign the act of sale as long as a permit faces an appeal. Developers thus have to wait for the case to be resolved to acquire a parcel and start building (Vautrin, 2018). Three successive reports by the judicial administration (Labetoulle, 2013; Maugüé, 2018; Pelletier, 2005) prompted reforms intended to protect authorized construction from abusive or opportunistic appeals while maintaining citizens' "access to the judge". Public debate surrounding this issue lacks empirical elements as official statistics are limited to the broad number of land-use litigation cases. The dataset we present in the following section addresses this shortcoming.

2.2 Construction of the litigation data

To measure litigation directly, we assemble a new dataset of over 200,000 land-use permit decisions from the archives of France's supreme administrative court, covering cases filed between 2005 and 2023. We convert the full text of each decision into variables recording information about the case, such as who brought the appeal, what was appealed, and how the court ruled.

To do so, we first segmented the text into four sections describing respectively the *Header*, the *Statement of facts*, the *Court's reasoning*, and the *Outcome* of the case. For each case, we extracted a set of 20 variables, using regular expressions for the most basic and structured information such as court names, and relying on an LLM (`Mistral-small-2503`) for more complex ones. In accordance with best practices described in Dell (2025), we used 350 hand-coded files to improve and validate our methodology. 10 cases were used as examples in the prompts to improve performance ("few-shot prompting"), 40 cases were used as a training set for prompt engineering and the remaining 300 were used as a validation set using standard evaluation metrics (Törnberg, 2024). Appendix C contains extensive details about our methodology.

Table 1 describes the variables extracted. The LLM consistently achieved high performance, with accuracy rates exceeding 90% for most key variables and above 95% for more than half of the variables. Accuracy fell to between 83% and 87% for three variables whose information is described in less structured ways: the address of the appealed project, the type of construction, and, for housing projects, the number of units. These scores significantly outperform previous similar exercises (Leek et al., 2024; Pelaez et al., 2024). We attribute this strong performance to the well-structured nature of judicial decision texts, which made few-shot prompting particularly effective. Appendix D details the measurement of extraction performance. Despite the LLM's strong overall performance, some extraction errors remain and are more

difficult to measure for rare categories. We mitigate this concern by focusing on categories of litigation with large numbers of observations such as building permits.

2.3 Complementary data

To complement our main dataset with other data sources, we match the names of the municipalities identified in the cases to their official municipal codes. There are approximately 35,000 municipalities in France, many with unusual spelling and frequent homonyms. To minimize errors, we used a conservative deterministic matching strategy that matched each name within the jurisdiction of the court where the case was held and left cases without a unique match as unresolved. Our method delivers a matching rate of 95.2%, and we exclude the unmatched observations from our analysis. Details can be found in Appendix C.5.1.

The matching procedure allows us to enrich the dataset with a very granular set of municipal characteristics over the 1999-2024 period. These additional variables cover population density, socio-economic characteristics, housing stock, land characteristics, environmental amenities, housing transactions and prices, and building permits. We describe the sources in detail in Appendix C.5.3.

3 AN ANATOMY OF LAND-USE LITIGATION

Figure 1 presents court cases recorded between 2005 and 2023 from the category of litigant (left node) through the object of complaint (middle nodes) to the outcome in court (right node). Nearly three quarters of cases are brought by individuals, sometimes acting through family real estate holdings (*SCI*). Companies and Associations are the other two main categories of litigants (12% and 8%) and public authorities account for the residual 8%[5].

The main motives for litigation are building permits and related development authorizations by a wide margin (79%). The blue flow represents *Anti-construction* cases, mostly litigation of granted permits, which seek to prevent construction and account for 72% of individuals' building permit cases. The red flow represents *Pro-construction* cases, which aim to enable construction by overturning an initial permit refusal or the downzoning of a parcel. Companies mostly file *Pro-construction* appeals: nearly 90% of their cases involve building permits of which 60% were initially refused. Associations litigate against granted building permits or planning documents, generally to preserve land.

Just over a quarter of cases are withdrawn or settled before the hearing, mainly involving *Anti-construction* cases. Among those that remain, 32% succeed on average in overturning the decision appealed.

About 60% of building-related cases target housing (Figure 1b). Among *Anti-construction* cases, multi-family (MF) housing represents more than 40% of cases compared to about 7% of authorized housing permits. Large infrastructure projects such as warehouses, energy (windmills, biogas plants) or telecommunication antennas that are frequently associated with NIMBYism (Jarvis, 2024) only represent about 7% of projects opposed. Litigation against construction thus appears mostly used by individuals against a mass of small projects, most often related to housing. This depicts a predominance of the routine NIMBY behavior, where residents oppose nearby development to protect local amenities, rather than targeted opposition to large projects typical of environmental conflicts (Rea et al., 2024).

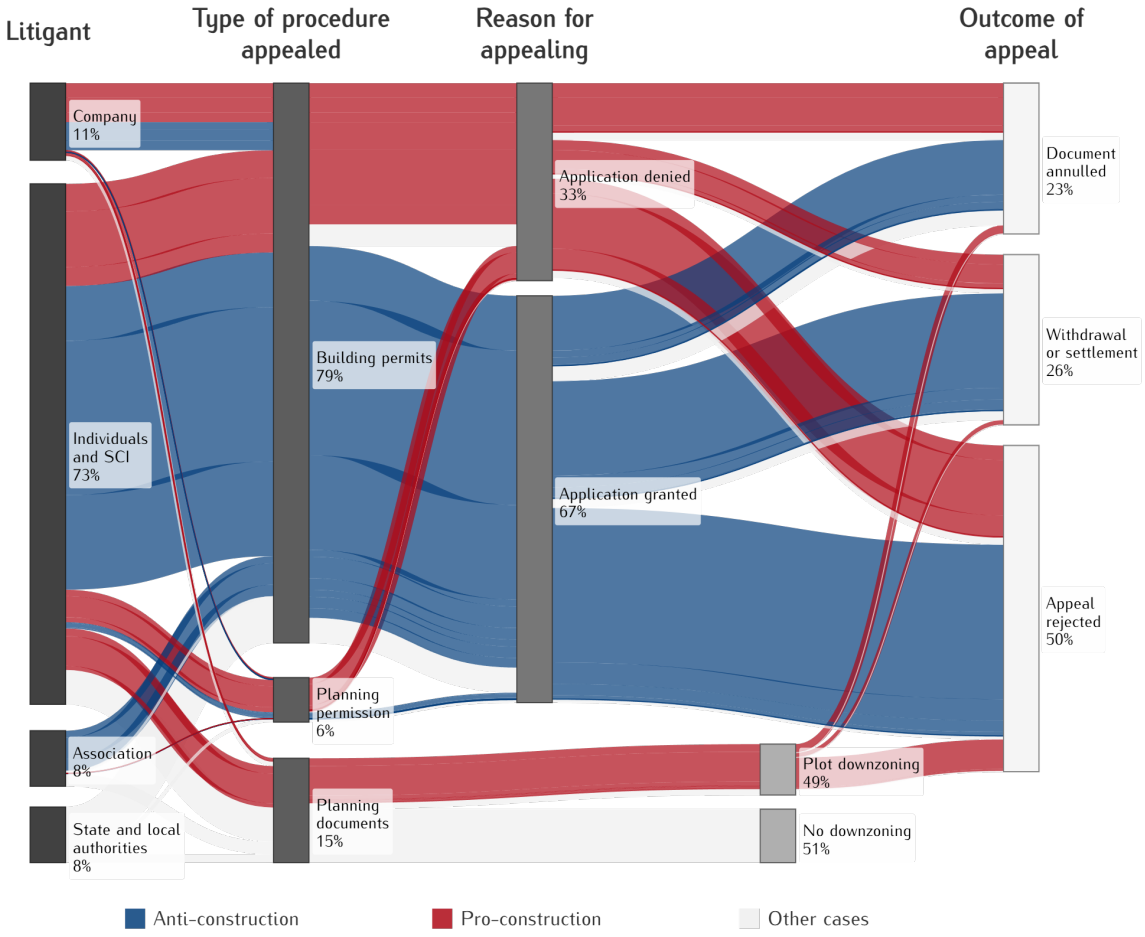
In the remainder of the article, we focus on litigation related to housing permits, which is the most frequent type of appeal.

5 Associations are mostly environmental NGOs opposing large projects or neighborhood committees representing homeowners. Public authorities (usually the *Préfet*, local State representatives, or mayors) often use courts to enforce public decision.

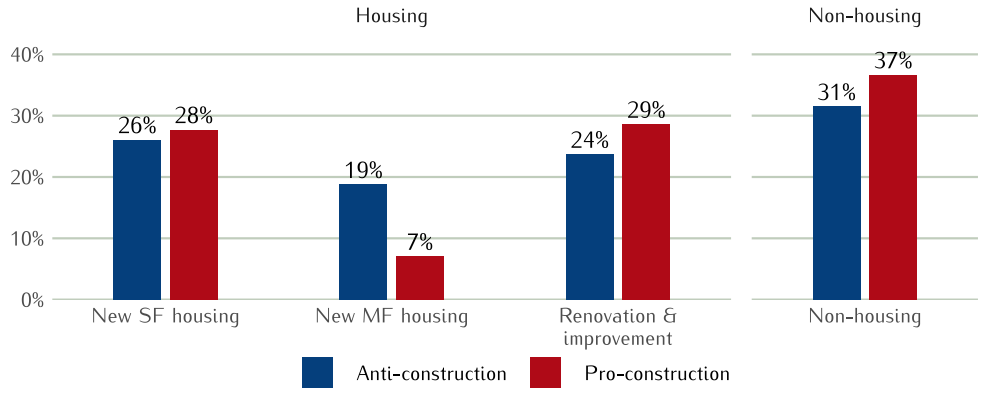
Item described	Variable	Type	Description	% NA	Accuracy
Timeframe	`date_application`	date	Date when the legal petition was filed with the administrative court	0%	99.3%
	`date_hearing`	date	Date of the hearing at the administrative court	0%	98.3%
Location	`court`	factor	City where the administrative tribunal is located (France has 42 administrative tribunals as of 2025)	0%	98.7%
	`operation_municipality`	string	Municipality where the project takes place or which is affected by the urban planning document in question	0.4%	96.3%
	`operation_address`	string	Specific address of the project, when the case involves a particular development project	46.2%	82.7%
Object of conflict	`administrative_act`	factor	Category of legal document contested in the petition (e.g., building permit, local urban planning scheme)	0.1%	93.0%
	`initial_outcome`	factor	Initial status of the legal act prior to the tribunal challenge (e.g., permit granted to neighbor, permit denied to applicant)	3.2%	94.3%
	`plot_classification`	logical	Dummy variable indicating whether the case involves reclassifying a parcel in the local urban planning scheme	0%	93.7%
Applicant	`applicant_type`	factor	Category of entity filing the petition (individual, company, association, etc.)	0.1%	95.7%
	`applicant_municipality`	string	Municipality where the petitioner resides	49.1%	98.3%
	`applicant_address`	string	Address of the petitioner when available in the document	50.3%	97.7%
	`lawyer`	logical	Dummy variable indicating whether the petitioner used a lawyer's services	0.2%	97.3%
Project	`housing`	logical	Dummy variable indicating whether the case involves residential housing	19.8%	94.0%
	`housing_number`	integer	Total number of residential units involved in the case	0.1%	87.4%
	`social_housing`	logical	Dummy variable indicating whether the case involves social housing	23.2%	99.3%
	`social_housing_number`	integer	Number of social housing units involved in the case	0.4%	100.0%
	`construction_type`	factor	Category of construction involved when the dispute concerns a specific project	20.7%	83.3%
Outcome	`outcome`	factor	Final ruling issued following the hearing	1.5%	95.7%
	`compensation`	integer	Financial compensation to be paid by the losing party to the winning party	0%	98.7%
	`irreducible_costs`	integer	Legal fees and court costs to be paid by the losing party to the winning party	0%	97.0%

Table 1: Description of variables extracted

Notes: For certain variables, the information is not systematically present in the text (e.g., petitioner's address or type of construction), for others, it is not applicable (number of dwellings if the case concerns a town planning scheme rather than a specific project). The list of categories for categorical variables can be found in the appendix. Table C.7 and Table C.6 in the appendix provide descriptive statistics of the variables.



(a) Litigant, procedure and outcome



(b) Type of project

Figure 1: Categories of cases in the Litigation database

Reading note: 73% of litigants are individuals or small family-holdings (*SCI*). 79% of litigation cases were against building permits. Individuals and SCIs challenging granted building permits represent 41% of all cases. In 49% of cases, planning documents face litigation because they have downzoned a parcel of land.

Note: The category 'Building permits' gathers the 3 types of building authorizations: building and demolition permits (often tied), development permits and prior notice (in housing often corresponding to renovation). Planning permissions are authorizations to build at a later date under the current land-use regulations. Excluded from the figure are cases concerning only financial compensation for claimants, along with cases related to land use that could not be readily classified (e.g., refusals to connect a building to the municipal water network or requests for exemption from fees associated with a public project). These categories account for 2.7% of cases in the database. Panel (b): shares of each type of project among building-permit cases, for *Anti-* and *Pro-construction* litigation respectively; the category 'Housing - other' (residences, care homes) is excluded (1.6% of *Anti* and 0.8% of *Pro* cases) and shares are computed over the four remaining categories ($N = 75,215$ permits for *Anti* and $N = 28,551$ for *Pro*).

4 A SIMPLE MODEL OF LITIGATION

This section sets out a simple model linking the probability that a building permit is appealed in court to the incentives of those entitled to challenge it. We start from the standard litigation model of Shavell (1982), in which a party sues when her expected gain exceeds her cost, and adapt it to land-use litigation in two ways. First, a neighbor who appeals against a permit seeks to prevent a loss rather than to recover one, and because the project is in practice frozen while the case is pending, delay has value in itself. In our data, the average time for a case to be judged in court is a little over 16 months. Second, contrary to the canonical model, the stake of the case is asymmetric in construction litigation: the developer defends the profit of its project whereas the neighbor defends her living environment, capitalized into the price of her house. The framework yields three predictions on how litigation varies with housing prices, project size and population density.

The payoff for an individual third party i filing against construction project j can be stated as:

$$\pi_{ij} = \left[q_j + (1 - q_j)\, g(T)\right] B_{ij} \tag{1}$$

where $B_{ij} = V_i\, \ell(\Delta z_{ij})$ with $\ell(0) = 0$, $\ell' > 0$. V_i is the value of her property and Δz_{ij} the loss of amenities from the project (an open view, low density, etc.), which fades with distance to the project. In defining B_{ij} this way we assume that a construction project has a larger impact in wealthy areas. This assumption follows capitalization studies, which find that the negative effects of nearby construction increase with the size of the project and tend to be larger in affluent, low density neighborhoods (Diamond & McQuade, 2019; Gyourko & McCulloch, 2024). An appeal thus pays off through two channels: delay, which secures a share $g(T)$ of the stake, with $g(0) = 0$ and $g(T) \to 1$ as the delay T grows; and annulment, which occurs with probability q_j and secures the rest.

Legal costs have two components: a one-time cost of filing c_i and an additional cost award c_i^L, which the judge may order the losing party to pay the winner. A litigant who wins with probability q_j therefore expects to pay $c_i + (1 - q_j)\, c_i^L$. In practice, legal cost compensation ($\approx$ 2,000€) is rarely granted and, when awarded, falls well below typical lawyers fees (from 5,000€ to 15,000€), making c_i^L negligible. A litigant can hire a lawyer which increases both c_i and the probability of winning q_j (Figure A.2, Appendix E.2). In the data, 72.5% of cases against approved construction involve a lawyer.

Letting a case run costs the developer

$$M_j = \left[q_j + (1 - q_j)\, \gamma(T)\right] \Pi_j, \tag{2}$$

where Π_j is the profit at stake, which increases with the size of the project and the local price per square meter, and $\gamma(T)$ the share of this loss incurred through delay[6]. The developer is therefore willing to pay up to M_j to have the appeal withdrawn (Appendix E.1). A litigant accepts the settlement amount if it is equal to the expected value of winning the case minus the legal costs. Thus, the filing condition becomes:

$$c_i \leq \max(\pi_{ij},\, M_j) \tag{3}$$

This implies that appeal can be worthwhile even when $\pi_{ij} < c_i$ (Rosenberg & Shavell, 1985). Withdrawals, the observable trace of settlements, account for 26% of outcomes in appeals to granted permits (Figure 1).

6 Defense costs add little, since the municipality, as author of the permit, usually defends it in court.

To aggregate individual probabilities of litigation p_{ij} at the project scale, one may assume like Cuttner et al. (2024) that they are independent. In that case, $p_{ij} = \mathbb{P}\left(c_i \leq \max(\pi_{ij},\, M_j)\right)$, and the project P_j is litigated if at least one of its N_j neighbors with standing does:

$$P_j = 1 - \prod_{i=1}^{N_j} \left(1 - p_{ij}\right), \tag{4}$$

and P_j rises mechanically with density through N_j. Given that an annulment protects every neighbor while the litigant alone bears its cost, each may wait for another to act, leading to fewer cases at high density levels (Olson, 1971). Associations are one way around the collective action problem: they file when the pooled stakes of their members cover the costs of litigating and organizing (subsection E.3). We indeed observe that the share of neighbor collectives among litigants rises to about 8% in the densest communes while that of environmental associations opposing land development is highest in low density areas (Figure A.3).

The same logic applies when a developer contests a refusal, with the crucial difference that only the applicant has standing, as there is no third-party appeal. The probability of litigation reduces to $\mathbb{P}\left(c_j \leq \tilde{q}_j\, \Pi_j\right)$, with $\tilde{q}_j$ the probability of overturning the refusal, and involves no aggregation over neighbors. Density matters only through the value of the project, and since delay now costs the litigant, there is no rent from holding out. Since Π_j is typically larger than c_j, even a relatively small $\tilde{q}_j$ makes an appeal worthwhile. Taken together, the framework predicts that *Anti-construction* litigation (i) rises with nearby property values through the appellant's expected payoff from winning, π_{ij}, (ii) rises with project size through the developer's willingness to pay for a withdrawal, M_j, and (iii) rises with density and the number of affected third parties. *Pro-construction* litigation, by contrast, should depend only on the project stakes rather than on the number of neighbors.

Housing and land use are, however, notably contentious issues, with high transaction costs and low willingness to bargain (Fischel, 2015, ch. 6). How much of the observed litigation reflects economic rationality rather than social norms or other noneconomic motives remains an open question (Ellickson, 1998), which we address in the next section.

5 EMPIRICAL ANALYSIS OF LITIGATION

The objective of this section is to assess the theoretical predictions that NIMBY behavior is driven by the combination of high housing prices and a dense environment in which each project affects several neighbors. We first build two metrics for each type of litigation $k \in \{\text{Anti}, \text{Pro}\}$: $\widehat{P}^k$ is the local aggregated litigation rate, the empirical counterpart of $\frac{1}{J}\sum_j P_j$ when there are J projects, and $\widehat{P}^k_U$ is the unit litigation rate. Our administrative data cover the universe of building permits in France and allow us to distinguish between granted and denied permits. We therefore define per municipality i:

$$\widehat{P}^{\text{Anti}} = \frac{\text{Appealed granted permits}}{\text{Granted permits}}, \qquad \widehat{P}^{\text{Anti}}_U = \frac{\text{Housing units in appealed granted permits}}{\text{Housing units in granted permits}}.$$

The *Pro-construction* rates $\widehat{P}^{\text{Pro}}$ and $\widehat{P}^{\text{Pro}}_U$ are defined analogously, using denied permits. Both measures compare aggregate litigation counts with aggregate permit counts within the municipality, rather than relying on litigation-to-permit matches. They should thus be read as area-level litigation rates rather than the probability that a given permit is appealed.

We apply several data-cleaning procedures that likely render our measures conservative. First, about 5% of cases could not be matched to a municipality and are thus excluded. Second, to avoid double counting, we remove litigation cases that may correspond to the same permit, often large projects appealed by several parties (Appendix C.6.1)[7]. Third, when the number of housing units is missing (about 10% of cases), we assign 1 unit to houses and the local average of permitted units to apartment buildings, which is conservative given that litigated permits tend to be denser than the average (about 29 units versus 22 nationwide). Finally, we winsorize the number of units at the 1st and 99th percentiles.

For $\widehat{P}^{Anti}$, the resulting sample includes 37,328 cases in mainland France between 2005 and 2022, representing 1.3% of granted permits and about 7.6% of authorized housing units. For *Pro-construction* cases, we keep 11,466 cases, $\widehat{P}^{Pro}$ equals 1.9% and $\widehat{P}_U^{Pro}$ about 4.5%, which suggests that the size of the permit is less of a determinant.

5.1 Spatial patterns of litigation

Figure 2 shows that $\widehat{P}_U^{Anti}$ and $\widehat{P}_U^{Pro}$ have different spatial patterns. $\widehat{P}_U^{Anti}$ is strongly concentrated around large metropolitan areas, on the Mediterranean and Atlantic coasts and in the Alps. *Pro-construction* litigation, by contrast, is more scattered across space. Moran's I measure of spatial autocorrelation calculated on commune litigation rates within distance rings shows that for $\widehat{P}_U^{Anti}$, it falls from 0.235 among communes less than 5 km apart and disappears beyond 75 km. Spatial dependence is therefore concentrated within roughly 30 to 50km, a distance comparable to that of a metropolitan area. For $\widehat{P}_U^{Pro}$ it never exceeds 0.040, indicating no spatial dependence. Panel 2c plots $\widehat{P}_U^{Anti}$ and $\widehat{P}_U^{Pro}$ along percentiles of household density. Both measures are approximately equivalent until the 75th percentile where $\widehat{P}_U^{Anti}$ rises much faster with density and is increasingly larger than $\widehat{P}_U^{Pro}$.

This provides an illustration of Hilber and Robert-Nicoud (2013)'s "influential landowner hypothesis" in which owners of developed land lobby for NIMBY-like restrictions that raise the value of their parcels whereas owners of undeveloped land lobby for development rights. As density increases, the former group dominates the latter in the political game.[8] Descriptively, these patterns are consistent with our theoretical framework, which predicts density and high housing prices to be primary determinants of *Anti-construction* litigation but says less about *Pro-construction* litigation. Since cases against refused permits are brought by applicants and not by third parties, density is unlikely to matter, although prices may: where they are high, a developer whose permit is denied loses more revenue. *Pro-construction* cases also provide a useful benchmark: if *Anti-construction* litigation simply reflected a general tendency to challenge local decisions, both types of cases should exhibit similar patterns. We test these predictions in the next section.

5.2 Local determinants of litigation

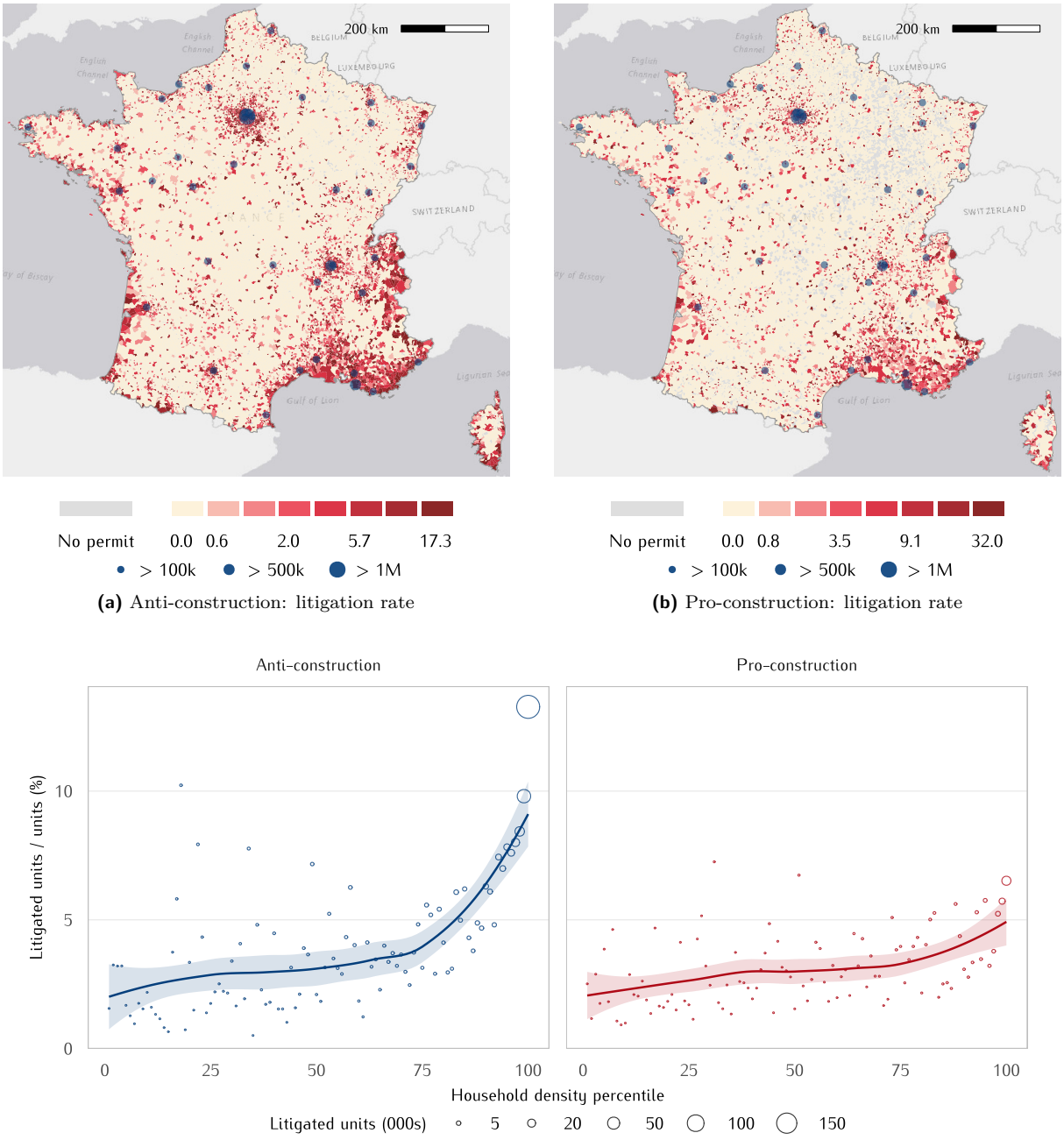
In this section, we estimate the local determinants of $\widehat{P}_U^{\text{Anti}}$ and compare them against those of $\widehat{P}_U^{\text{Pro}}$. We restrict the analysis to the 682 French Functional Urban Areas (FUAs), leaving out rural municipalities, where both construction and litigation are rare.[9]

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7 We apply the same cleaning procedures for $\widehat{P}^{Pro}$ although occurrences of multiple cases for a same permit are much rarer.

8 Our measure understates *Pro-construction* litigation for low-density centiles as it leaves out landowners' appeals against plot downzoning. These tend to be located at the urban fringe as buildable parcels fetch a very large premium compared to agricultural land: in 2024, agricultural land sold for 6,400€/ha on average, against 377,000€/ha for buildable land (Safer, 2025). The cancellation of building rights therefore represents a significant loss of land rent estimated at about 4 years of median income in the municipalities of the impacted landowners.

9 Rural municipalities number nearly 9,000 ($\approx$ 26% of the total) but account for only 7% of authorized housing units and about 3.5% of litigated units over our period.



(a) Anti-construction: litigation rate

(b) Pro-construction: litigation rate

(c) Litigation and density

Figure 2: Litigation rates across space and density

Note: Top panels: litigation rate per authorized housing unit (left, anti-construction) and per denied housing unit (right, pro-construction), aggregated at the municipal level over the period and shown in quantile classes; municipalities in gray record no permit of the relevant kind. Blue dots represent the main cities, sized by population. Bottom panels: litigation rate by percentile bin of household density, the line is a local polynomial regression of degree 2 of the percentile rates on the percentile rank displayed with the 95% confidence interval of the fit. **Reading note:** Anti-construction litigation rates are highest in dense urban areas, on the coast and in the Alps, and rise sharply above the 75th percentile of household density. Pro-construction litigation is more evenly spread across the territory and its gradient is flatter and continuous.

$\widehat{P}_U^{\text{Anti}}$ varies substantially both across and within FUAs, in particular in small FUAs (below 200,000 inhabitants) where it reaches very high-levels in prized locations such as Monaco (38%) (Table A.1).

We estimate the following regression at the municipal level:

$$\widehat{P}_i^k = \beta_0 + \beta_1 \,\text{Price}_i + \beta_2 \,\text{Density}_i + \beta_3 \,\text{Income}_i + \beta_4 \,\text{MFHousing}_i + \boldsymbol{\gamma}^\top \mathbf{X}_i + \varepsilon_i^k, \tag{5}$$

where i denotes the municipality and $k \in \{\text{Anti}, \text{Pro}\}$ the type of litigation. Price_i is the mean house price per square meter observed in local transactions, Density_i the number of households per square kilometer, Income_i the average income per adult, and MFHousing_i the share of multifamily housing in authorized units. The vector $\mathbf{X}_i$ contains the following controls: the share of homeowners, the shares of blue- and white-collar workers, distance to the coast, the share of secondary homes in the housing stock, a construction-pressure index (new authorizations relative to the 2006 housing stock), the share of recreational areas within municipality and distance to the nearest administrative court. It also includes FUA fixed effects, so the coefficients are identified from variation across municipalities within the same FUA. All control variables are averaged over the sample period. Following Millar et al. (2016), we center all continuous variables by subtracting their sample medians. The constant is therefore interpreted as the predicted litigation rate for a *baseline municipality*, and belonging to the omitted city, which has the litigation rate closest to the average for large (resp. small) cities.

Our total sample includes about 22,500 municipalities, with an average size of 2,300 inhabitants. We estimate the model first for all FUAs, and then separately for large and smaller FUAs above the median of the new housing supply distribution. Following Barza et al. (2024), we estimate Equation 5 weighting municipalities by the number of authorized permits. Litigation rates based on more permits are measured more precisely, and weighting captures litigation risk for the average permitted project. In Appendix B, we show that results are robust to using non-weighted regression, changing the outcome variable for its unit-weighted version $\widehat{P}_U$, and using an alternative set of controls.

Table 2 presents the estimates with standard errors clustered at the FUA level. Across all specifications, the constant term is not statistically different from zero. Litigation against construction thus does not seem to be an important issue for a typical municipality. Starting with $\widehat{P}^{\text{Anti}}$ in columns (1) to (3) , results are consistent with our litigation model. The strongest predictor overall is housing prices. The effect is sizeable: moving from the median value to the 90th percentile of municipal prices per square meter increases the litigation rate by 1.07 percentage points (+137%), 1.1 pp (+84%) and +0.7 pp (+116%) in all, large and small FUAs respectively. In large FUAs (2), the second most important variables are household density, mean income and the share of multi-family housing in new construction: moving from the median value to the 90th percentile increases $\widehat{P}^{\text{Anti}}$ respectively by 62%, 59% and 37%. In smaller FUAs (3), density is not significant and the share of secondary homes is the second largest predictor (+90%) followed by the share of MF housing (+57%). We return to this result below.

Interestingly, the coefficient on the homeownership rate is small and statistically insignificant. This result is robust to the inclusion of age bands to the regression and their interaction with the homeownership rate (see Appendix A.3). This does not imply that renters litigate as much as homeowners, since the appeal process favors nearby owners. Rather, it suggests that appeals depend less on the overall prevalence of homeownership than on whether local owners have the financial incentives and means to challenge development.

Regarding $\widehat{P}^{\text{Pro}}$ in column (4), the coefficient on household density is only significant at the 10% level and is comparatively 7 times smaller than in column (1): moving from the median value to its 90th percentile increases $\widehat{P}^{\text{Pro}}$ by 10% (versus 70% in column (1)). Price is also relatively less important, a similar movement along the distribution only increases $\widehat{P}^{\text{Pro}}$ by 45%. More generally, the model has substantially less explanatory power for $\widehat{P}^{\text{Pro}}$, with an R^2 of 18% versus above 50% for $\widehat{P}^{\text{Anti}}$. Separate results for large and small FUAs are qualitatively similar (Table A.2).

	Anti-construction			Pro-construction
	All FUAs	Big FUAs	Small FUAs	All FUAs
(Intercept)	−0.613	−0.107	−0.159	−0.588
	(0.509)	(0.548)	(0.343)	(0.655)
Mean price per square meter	0.114***	0.096***	0.114***	0.071***
	(0.018)	(0.023)	(0.020)	(0.017)
Household density	0.509***	0.518***	0.177	0.113***
	(0.034)	(0.041)	(0.114)	(0.042)
Mean income per adult	0.067*	0.106**	0.030	−0.010
	(0.038)	(0.048)	(0.030)	(0.020)
Share MF housing	0.735***	0.898***	1.075***	0.610**
	(0.189)	(0.223)	(0.231)	(0.284)
Homeowners	0.215	−0.804	0.332	1.731***
	(0.489)	(0.625)	(0.568)	(0.459)
Blue collar	0.026**	0.037**	0.011	−0.001
	(0.011)	(0.017)	(0.007)	(0.013)
White collar	0.020	0.028	−0.036	0.040
	(0.020)	(0.021)	(0.025)	(0.029)
Secondary homes	3.244***	3.578***	2.441***	2.523***
	(0.475)	(0.927)	(0.494)	(0.657)
Dist. to coast	0.006**	0.007**	0.003	0.005
	(0.003)	(0.003)	(0.003)	(0.005)
Units authorized / stock	−1.078***	−1.145***	−0.969***	−1.390***
	(0.247)	(0.302)	(0.205)	(0.338)
Recreational area	0.000	0.001	−0.001	0.003
	(0.002)	(0.003)	(0.002)	(0.004)
Dist. to court	−0.001	0.000	0.003	−0.009**
	(0.003)	(0.004)	(0.002)	(0.004)
Observations	22,534	8,864	10,577	21,103
Adj. R^2	0.587	0.606	0.502	0.182
Fixed Effects: FUA	Yes	Yes	Yes	Yes
Weights	Permits	Permits	Permits	Denied permits

Table 2: Determinants of the litigation rate

Notes: This table presents coefficients from the OLS regressions at the municipality level. The dependent variable is the litigation rate (share of building permits that are litigated). Columns report results for all FUAs, large FUAs, and small FUAs; columns with "Interc." include an interaction between the homeowners' share and household density. All specifications include FUA fixed effects and the full set of controls described in the text; continuous covariates are median–centered. Standard errors, clustered at the FUA level, are reported in parentheses. **Signif.:** $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Taken together, these results corroborate our theoretical predictions that opposition to construction is more likely in wealthy and dense areas and when new housing production takes the form of multi-family housing. The positive effect of income, combined with the propensity of individuals to hire a lawyer further suggests that litigation originates from high-income individuals. These associations should not be interpreted causally, however. Since the number of units authorized relative to the 2006 housing stock is negatively associated with litigation, litigation could itself raise housing prices by making supply less elastic. To address this potential reverse causality, Appendix B.1 presents a panel specification. Estimates are noisier in small cities, where construction and litigation fluctuate more across periods, but they confirm the central role of housing prices and multi-family housing everywhere, and of density in large cities. The share of secondary homes in small cities loses significance, possibly because it varies little over time. The final section of the empirical analysis examines how the determinants of $\widehat{P}^{\text{Anti}}$ differ between large and small FUAs, with particular attention to secondary housing.

5.3 Secondary homes and absentee litigants

To compare the relative importance of each group of determinants of $\widehat{P}^{\text{Anti}}$ in large and small FUAs, we use a Shapley decomposition (Shorrocks, 2013). The contribution of each group is its incremental R^2 when added to the model, averaged over all possible orderings in which the groups could enter, so that the result does not depend on entry order. Shares are expressed as a percentage of the explained variance. FUA fixed effects are included in all specifications, so the results capture within-FUA variation only. Consistent with the results in the previous section, Figure 3a shows that household density and prices explain most of the variance in large FUAs while homeowners and the socio-demographic characteristics explain little. In small FUAs, the share of secondary homes comes just after housing supply (share of MF housing and our construction index), whereas it matters very little in large FUAs.

We consider two possible explanations for the importance of secondary homes. First, local residents may oppose new construction more where secondary homes are widespread, because new housing is likely to be bought by non-residents and crowds out supply for the local population. Second, owners of secondary homes may themselves litigate against new projects to preserve the pristine environment of their holiday homes. To test these hypotheses, we use the fact that, until 2013, hearing transcripts recorded litigants' addresses. For 2005-2012, we can therefore distinguish litigants who live in the municipality where the appealed permit was granted from those who live elsewhere, which we call "absentee litigants". Many of them are likely to be owners of secondary homes: they account for more than half of litigated permits in Saint-Raphaël or Chamonix[10].

To quantify their role in local appeals, we first rerun the Shapley decomposition with and without cases brought by absentee litigants. If owners of secondary homes drive part of the litigation, the share of variance explained by secondary homes should fall once their cases are removed. Figure 3a confirms this. In small FUAs, the share explained by secondary homes drops from 21.3% to 15.4% and that of prices from 21.3% to 16.8%, while the share of density rises from 13.8% to 18.7%. The decomposition of small FUAs without absentee litigant is thus relatively more similar to that of large FUAs, which are less affected by the share of secondary homes, due to housing markets being much larger there.

Second, we decompose the secondary-homes coefficient by litigant status. Each litigated permit is attributed to an absentee, a local, or a litigant with an unknown address, so the litigation rate is the sum of three status-specific rates. When each is regressed on the same specification as in Equation 5, the coefficients add up exactly, since weighted OLS is linear in the dependent variable: $\hat{\gamma}_{\text{SH}} =$

10 This classification is likely a lower bound. The address cannot be matched for 19% of litigated permits in small FUAs and 9% in large ones, and some owners may declare their secondary home as their residence, in which case they are classified as local.

$\hat{\gamma}_{SH}^{absentee} + \hat{\gamma}_{SH}^{local} + \hat{\gamma}_{SH}^{unknown}$. The absentee share $s^{abs} = \hat{\gamma}_{SH}^{absentee}/\hat{\gamma}_{SH}$ is thus the fraction of the coefficient that disappears without absentee cases; adding $s^{unknown}$ gives an upper bound, reached if all unknown addresses are absentee.

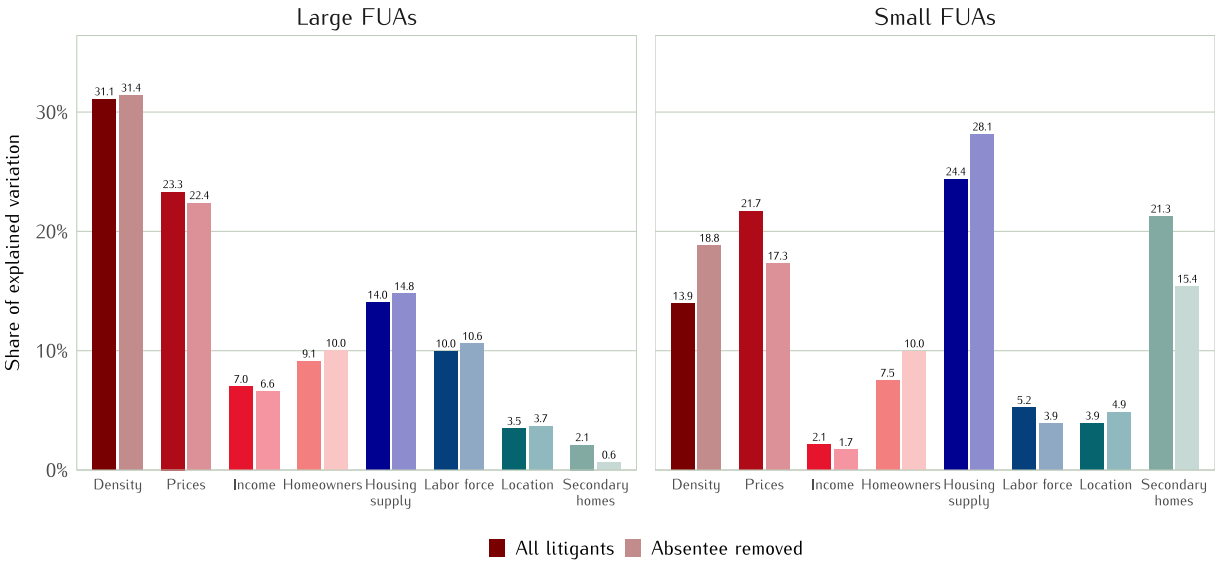
Absentee litigants account for 63.9% of the secondary homes coefficient in large FUAs and 38.7% in small FUAs (76.7% and 62.8% respectively under the upper bound). This drop is much larger than for any other determinant. In large FUAs, the effect of secondary homes disappears once absentee cases are removed. In small FUAs, it falls substantially, but local litigants still account for 37.2% of the coefficient, so we cannot rule out the crowding-out explanation. Together, these results suggest that much of the effect of secondary homes reflects litigation by their owners, who use local courts to keep new construction away from their holiday homes.

6 CONCLUDING REMARKS

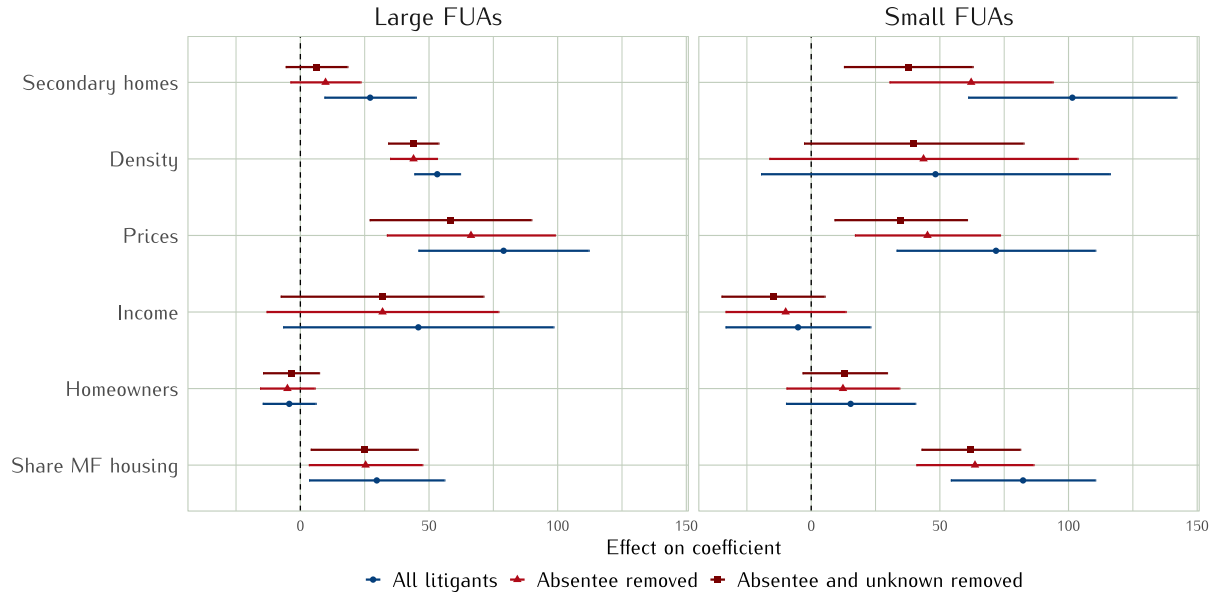
This paper uses a new dataset on litigation in France to provide direct evidence of NIMBY opposition. We document that the majority of court cases are brought by individuals appealing building permits against new housing, rather than against major projects commonly associated with environmental harm. To explain the local determinants of this routine NIMBYism, we develop a simple theoretical framework linking benefits and costs of litigation to the appeal rate. Consistent with its predictions, the appeal rate increases with local housing prices, density and the amount of multi-family housing built. In small urban areas, litigation is most strongly associated with the prevalence of secondary homes. This is driven by their owners: excluding "absentee litigants" significantly reduces the corresponding coefficient.

Not all land-use litigation opposes development: owners of undeveloped land challenge permit denials to defend their building rights. On average, they appeal more frequently than third parties challenge granted permits, although the latter becomes relatively more important as population density increases. These findings inform how competing interests shape urban development through the courts. This nuances the "homevoter hypothesis", underlining the role of wealthy owners, not the median homeowner.

Our findings also highlight potential implementation costs for policies that promote densification and prevent new greenfield development, especially when costs of appealing are low and delays associated are high. Unequal access to courts and lawyers may further imply that densification policies have adverse distributive consequences.



(a) Shapley decomposition of the within FUA variance



(b) Sensitivity of the litigation-rate determinants to absentee litigants

Figure 3: Comparison of local determinants of *Anti-construction* litigation with and without absentee litigants

Note (a): Shapley decomposition of the regression R^2, within FUA. Variables are grouped as follows. *Density*: household density (GHS built-up surface) and its square. *Prices*: mean price per square meter. *Income*: mean income per adult. *Homeowners*: share of homeowners. *Housing supply*: share of multi-family housing and authorized units over housing stock. *Labor force*: shares of white-collar and blue-collar workers. *Secondary homes*: share of secondary homes. *Location*: distance to the coast, recreational area and distance to the administrative court. Observations are weighted by authorized housing units.
Note (b): Each point is the estimated effect of moving a commune from the median (p50) to the 90th percentile (p90) of a determinant on the 2005–2012 litigation rate. It is expressed as a percentage of the mean litigation rate in the all-litigants sample, and bars show 95% confidence intervals, clustered by FUA. The litigation rate splits exactly into three parts: cases filed by confirmed absentee litigants, cases filed by confirmed local litigants, and cases filed by litigants whose address is unknown.

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A ADDITIONAL RESULTS

A.1 Temporal trends of land use litigation

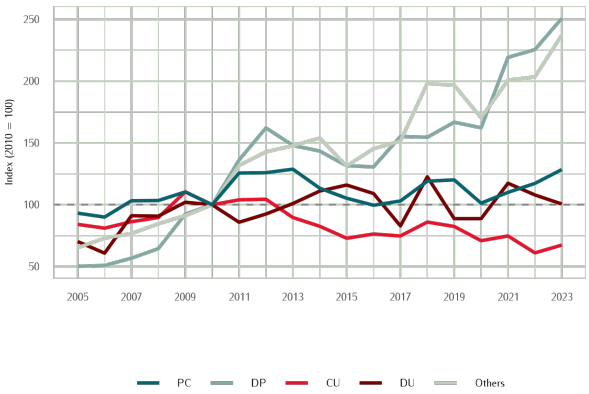
Absolute number of litigation cases, plotted in Figure A.1c, suggest that land-use related conflicts are on a slight upward trend in France. Since the early 2010s, annual growth in the number of cases has toned down relative to the 2005-2012 period, corresponding to reforms of Administrative Law limiting the ease with which one may litigate on land-use matters. Indeed, between 2009 and 2011 the flow of new land-use related cases before administrative increased by approximately 8% each year. The number stabilized between 2011 and 2013, followed by a decline in 2013, which corresponds to the implementation of the ordinance of July 18, 2013 (Légifrance, 2013), restricting the definition of interest in bringing proceedings (*intérêt à agir*) and therefore of the parties who may legitimately initiate urban planning litigation (Delesalle, 2016).

The increase is driven by a relatively larger rise of *Anti-construction* cases: +69% over the period compared to +17% for *Pro-construction* cases ("Other" decreased by 5%). Litigation against building permits increased by about 30% over the period when the number of permit applications oscillated around their 2005 level (Figure A.1b) - building permits exhibit annual oscillations due to cyclical economic condictions, policy adjustment and market demand fluctuation.

Cases related to planning documents increased by about 40% (Figure A.1a). Recently, a growing number of cases have been made against larger-scale, supra-municipal planning documents, involving dozens of municipalities. A simple trend comparison with the number of municipalities approving documents is only partially informative as many legal challenges to planning documents occur years after their approval. Nonetheless, the upward trend of cases against planning documents is at odds with this decrease and suggests an increase in the number of legal challenges per approved document that local governments may face when designing land use regulation. French legal scholars have argued that the growing complexity of urban-planning law, and its increasing overlap with environmental law, has created more grounds for litigation, as planning documents must now include explicit construction rules and environmental and social objectives (Delesalle, 2016).



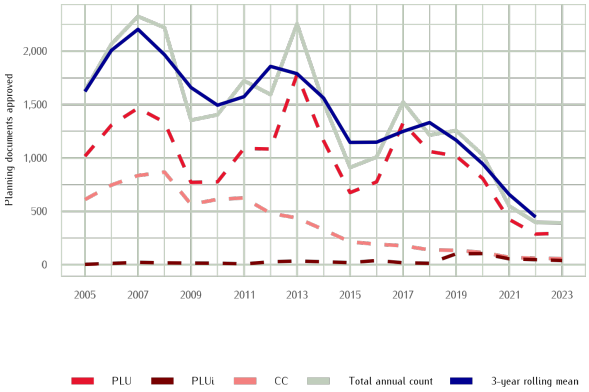
(a) Conflict type — base 100 (2010 = 100)



(b) Administrative act — base 100 (2010 = 100)



(c) Building permit applications — levels

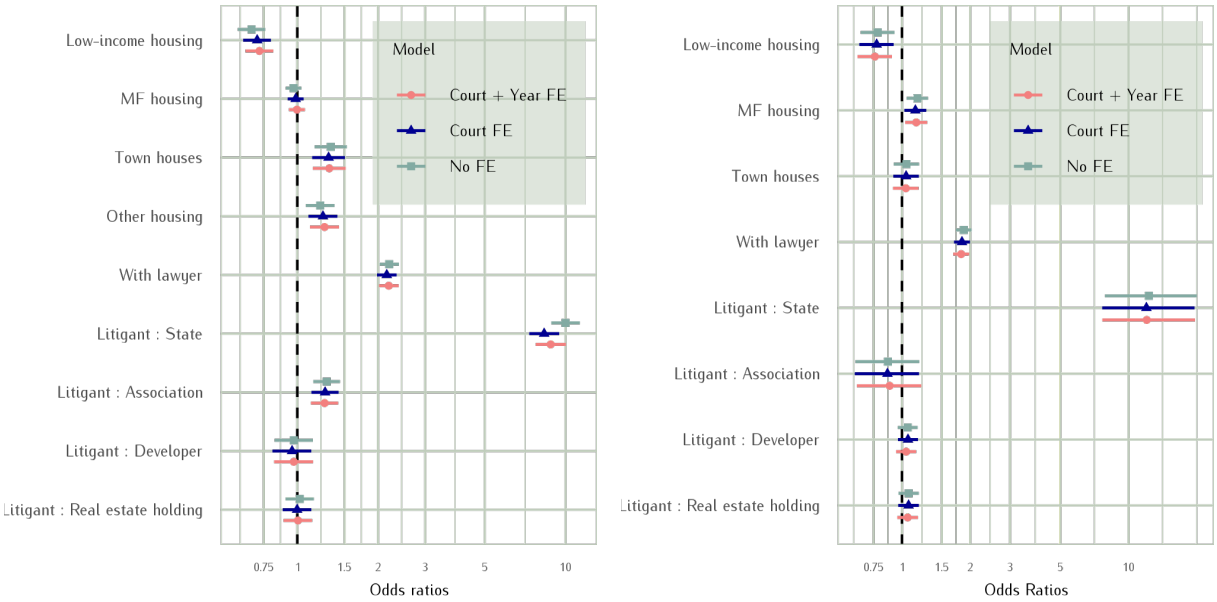


(d) Planning documents — levels

Figure A.1: Comparative evolution of litigation cases and administrative acts

Note: Cases are plotted by year of hearing. DU = planning documents; PC = building permits; CU = land-use certificates; DP = renovation and transformation permits. All housing-related building permits, whether granted or denied, are included. The recent decline in approved planning documents may reflect the fact that a larger share of municipalities already has a valid planning document.

A.2 Outcomes of litigation



(a) Cases against authorized building permits

(b) Cases against denied building permits

Figure A.2: Estimated coefficients from binomial logit models of case success

Note: Each panel shows three specifications: (1) baseline model with no fixed effects, (2) model with court fixed effects, and (3) model with court and hearing-year fixed effects. All specifications control for claimant type, legal representation, and housing-related case characteristics. 'Real estate holdings' are family-hold investment companies created for property management.

A.3 City-level litigation rates

Large FUAs						Small FUAs					
Rank	City	$\widehat{P}$	$\widehat{P}_U$	σ	σ_U	Rank	City	$\widehat{P}$	$\widehat{P}_U$	σ	σ_U
1	Cannes - Antibes	6.6	22.1	4.5	11.5	1	Val-d'Isère	52.4	69.1	–	–
2	Nice	5.7	17.5	5.1	10.0	2	Saint-Tropez	17.0	38.6	3.8	1.4
3	Toulon	4.8	17.4	3.9	11.8	3	Monaco - Menton	14.6	38.1	7.5	18.4
4	Marseille - Aix-en-Provence	4.7	21.4	3.5	15.0	4	Bormes-les-Mimosas	11.8	26.9	5.0	12.3
5	Paris	3.9	12.8	5.2	12.1	5	Ciboure	6.5	31.2	5.1	6.0
6	Bayonne	2.9	15.6	3.6	10.9	6	Quiberon	6.5	24.6	0.9	2.5
7	Avignon	2.6	9.8	2.3	8.3	7	Propriano	6.4	33.3	5.2	24.1
8	Grenoble	2.4	9.6	2.7	9.0	8	St-Rémy-de-Provence	6.3	54.4	2.4	14.3
9	Lyon	2.2	10.0	2.5	7.6	9	Hendaye	6.1	30.6	4.5	15.1
10	Annecy	1.9	8.8	2.7	11.3	10	Le Pouzin	5.9	6.5	–	–
⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮
41	Amiens	0.6	3.9	1.3	3.8	490	Tulle	0.0	0.0	0.0	0.0
42	Orléans	0.6	2.8	1.1	3.6	490	Varennes-sur-Allier	0.0	0.0	0.0	0.0
43	Angers	0.5	1.9	0.9	2.1	490	Vic-Fezensac	0.0	0.0	0.0	0.0
44	Troyes	0.5	1.5	1.1	2.0	490	Vierzon	0.0	0.0	0.0	0.0
45	Valenciennes	0.4	2.5	1.0	7.5	490	Villaines-la-Juhel	0.0	0.0	0.0	0.0
46	Poitiers	0.4	4.4	0.8	5.7	490	Villedieu-les-PR	0.0	0.0	0.0	0.0
47	Dunkerque	0.3	1.3	0.9	3.9	490	Villers-la-Montagne	0.0	0.0	–	–
48	Le Mans	0.3	2.2	0.6	3.0	490	Vittel - Contrexéville	0.0	0.0	0.0	0.0
49	Limoges	0.3	2.9	0.3	3.3	490	Égletons	0.0	0.0	0.0	0.0
50	Lens - Liévin	0.2	0.7	0.5	1.9	490	Étain	0.0	0.0	0.0	0.0

Table A.1: *Anti-construction* litigation rates per city

Notes: Cities correspond to Functional Urban Areas (FUA). L is defined as the sum of litigated permits in the area divided by the same of granted permits. U corresponds to the sum of litigated units divided by the sum of granted units. σ_L and σ_U indicate (weighted) standard deviations. Cities comprised of only one large municipality like Val-d'Isère or Le Pouzin do not have a value.

A.4 Pro-construction litigation

	Pro-construction		
	All FUAs	Big FUAs	Small FUAs
(Intercept)	−0.588	−0.309	0.314
	(0.655)	(0.886)	(0.523)
Mean price per square meter	0.071***	0.059***	0.090***
	(0.017)	(0.021)	(0.019)
Household density	0.113***	0.118**	0.015
	(0.042)	(0.049)	(0.134)
Mean income per adult	−0.010	0.010	−0.062**
	(0.020)	(0.023)	(0.030)
Share MF housing	0.610**	0.779**	0.445
	(0.284)	(0.356)	(0.426)
Homeowners	1.731***	1.371**	2.169***
	(0.459)	(0.623)	(0.782)
Blue collar	−0.001	0.008	−0.011
	(0.013)	(0.019)	(0.016)
White collar	0.040	0.033	0.082**
	(0.029)	(0.035)	(0.037)
Secondary homes	2.523***	3.109**	1.677***
	(0.657)	(1.220)	(0.638)
Dist. to coast	0.005	0.006	0.003
	(0.005)	(0.006)	(0.007)
Units authorized / stock	−1.390***	−1.425***	−1.367***
	(0.338)	(0.410)	(0.384)
Recreational area	0.003	0.005	0.002
	(0.004)	(0.007)	(0.003)
Dist. to court	−0.009**	−0.012***	0.004
	(0.004)	(0.005)	(0.005)
Observations	21,103	8,553	8,986
Adj. R^2	0.182	0.185	0.188
Fixed Effects: FUA	Yes	Yes	Yes
Weights	Denied permits	Denied permits	Denied permits

Table A.2: Determinants of the pro-construction litigation rate.

Note: Appeals against a refused permit, as a share of refused permits. Same specification as the anti-construction baseline, weighted by refused permits and clustered at FUA level. Refused permits are defined on the dossier status (`STA_COD_STATDOSS`).

A.5 Age of homeowners

	Anti-construction			
	Big FUAs	Big FUAs, interac.	Small FUAs	Small FUAs, interac.
(Intercept)	−0.045	−0.088	−0.163	−0.141
	(0.577)	(0.583)	(0.344)	(0.340)
Mean price per square meter	0.098***	0.100***	0.115***	0.113***
	(0.023)	(0.023)	(0.020)	(0.020)
Household density	0.483***	0.489***	0.170	0.170
	(0.038)	(0.036)	(0.114)	(0.115)
Mean income per adult	0.103**	0.099**	0.033	0.032
	(0.046)	(0.045)	(0.030)	(0.030)
Share MF housing	0.996***	0.935***	1.102***	1.058***
	(0.225)	(0.231)	(0.233)	(0.233)
Homeowners	0.182	0.179	0.570	0.556
	(0.809)	(0.890)	(0.496)	(0.499)
Blue collar	0.049**	0.047**	0.017**	0.016**
	(0.021)	(0.020)	(0.007)	(0.007)
White collar	0.064***	0.064***	−0.020	−0.020
	(0.021)	(0.021)	(0.021)	(0.021)
Secondary homes	2.815***	2.848***	2.232***	2.289***
	(0.919)	(0.909)	(0.481)	(0.483)
Dist. to coast	0.006**	0.007**	0.003	0.002
	(0.003)	(0.003)	(0.003)	(0.003)
Units authorized / stock	−1.188***	−1.157***	−1.020***	−0.982***
	(0.310)	(0.293)	(0.229)	(0.220)
Recreational area	0.001	0.001	−0.001	−0.001
	(0.003)	(0.003)	(0.002)	(0.002)
Dist. to court	0.000	−0.001	0.003	0.003
	(0.004)	(0.003)	(0.002)	(0.002)
Aged 0–29	−0.042	−0.033	−0.018	−0.012
	(0.055)	(0.051)	(0.020)	(0.020)
Aged 45–59	−0.168***	−0.161***	−0.051**	−0.044**
	(0.031)	(0.027)	(0.022)	(0.021)
Aged over 60	0.003	0.007	−0.003	0.000
	(0.033)	(0.032)	(0.014)	(0.014)
Homeowners × Aged over 60		−0.090		−0.058
		(0.103)		(0.043)
Observations	8,864	8,864	10,577	10,577
Fixed Effects: FUA	Yes	Yes	Yes	Yes
Adj. R^2	0.612	0.612	0.503	0.504

Table A.3: Determinants of the litigation rate: age structure.

Note: Reduced baseline specification with the shares of the municipal population aged 0 - 14, 15 - 29, 45 - 59, 60 - 74 and 75 and over (in %, census, averaged over 2005 - 2022); the share aged 30 - 44 is the reference. Columns "interac." add the interaction between the homeowners' share and the share aged 60–74. Continuous covariates are median-centered. Weighted by authorized permits, clustered at FUA level. For parsimony we only report interaction results for one age band, but none are statistically significant

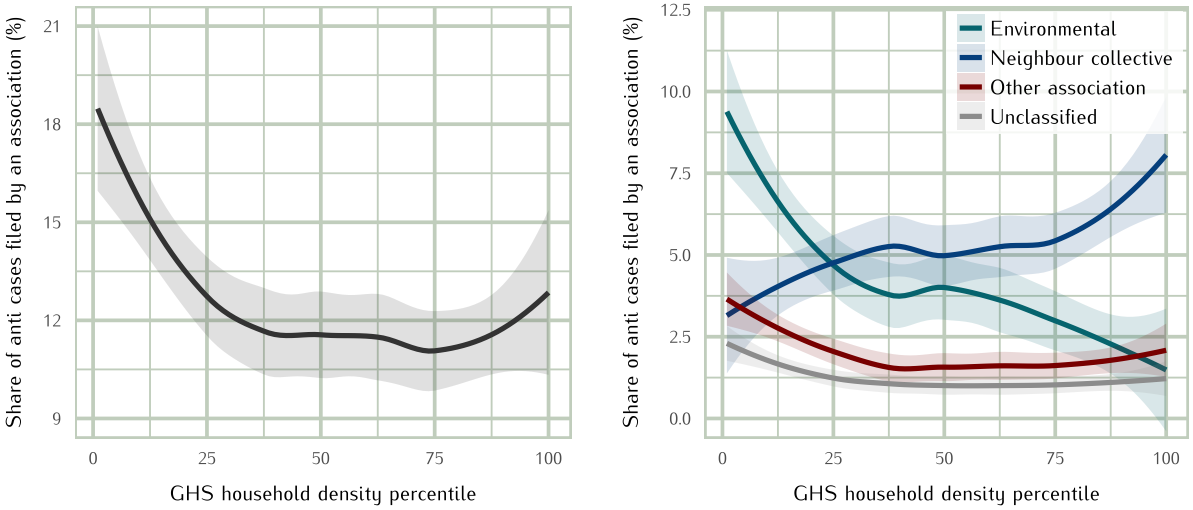
A.6 Litigation by associations

Since the "Association" category of the `application_type` variable includes groups with a variety of interests, we classify them according to their stated purpose and/or their name. Of the 17,494 cases in which an association (often several associations) is a plaintiff, an association name could be extracted from the text of the request in 99.4% of the cases, using the introductory keyword (*association*, *syndicat*, *collectif*, *comité*...). The registered address and ZIP code of the associations are available in 46% of cases. We match case-association pairs to the official registry of associations (*Répertoire National des Associations*, snapshot of 1 October 2026, 3.53 million entries) by exact normalized title, narrowed by postal code, commune and active status, and then by fuzzy matching within the commune. A registry entry is found for 54.3% of cases (51.4% of named associations). Unmatched names are classified from keywords in the name. Syndicates of co-owners (*syndicats de copropriétaires*, *conseils syndicaux*) and legal structures for joint land/property management,(ASL, AFUL, AFU, ASA) are assigned from their name alone. Overall, 51.0% of cases are classified from the registry, 19.9% from the name of the homeowners' association, and 25.5% from other name keywords.

For matched associations, the category is derived from the thematic code of the registry. The most frequent registry themes are environment and living environment (*environnement, cadre de vie*; 56% of matched associations), heritage preservation (*préservation du patrimoine*; 8%), societies and mutual-aid groups (*amicales et groupements d'entraide*; 6%), local development and solidarity (*développement local et solidarités*; 6%), defense of economic interests (*défense d'intérêts économiques*; 4%) and housing (*logement*; 4%). The registry pools environment and living environment in a single theme, so the most generic codes of this theme cannot distinguish nature protection from the defense of residents' interests. These generic codes are resolved from the stated purpose of the association. A purpose mentioning both themes is coded "Environment" only if it contains a marker specific to nature protection. Three categories result:

- Environment: nature protection associations (registry codes on pollution, nature, fauna, flora, ecology and sustainable development; keywords such as *biodiversité, faune, écologie, paysage, littoral*), 25.6% of cases;
- Neighbor collective: bodies defending the living environment of their members (*cadre de vie*, neighborhood committees, tenants and landowners, co-ownership syndicates and *associations syndicales*; keywords such as *riverains, habitants, quartier, copropriétaires*), 45.4% of cases;
- Other: all remaining associations, mainly heritage (36%), professional syndicates (14%) and religious or cultural bodies (13%), together with public bodies (*syndicats mixtes*, water and sanitation unions), 16.5% of cases.

Names that remain unclassified (no keyword or ambiguous match) account for 11.8% and cases without an identifiable name for 0.6%.



(a) All associations together

(b) Categories of associations

Figure A.3: Share of associations among litigants in *Anti-construction* cases by density

Note: Each point is the share of anti-construction cases filed by an association within one of 100 density percentile bins, cut on communes with at least one anti-construction case. Commune shares are averaged with weights equal to the number of cases, so a bin value is the share of cases filed in that bin. Curves are local polynomial (LOESS) regressions of degree 2 fitted on the 100 bin values, with a span of 0.75, meaning that each fitted value uses the nearest 75% of bins. Shaded areas show the 95% confidence interval of the smoothed curve. They reflect how well the curve fits the bin values and not the sampling error of each bin. No controls are included.

B ROBUSTNESS

All four variants drop the unemployment rate, the blue-collar share, the foreigner share and the Mountain Law indicator from the baseline covariates. Dropping them costs at most 0.003 of R^2 and leaves the price and density coefficients unchanged in sign, size and significance.

B.1 Panel, reduced covariates

	Anti-construction		
	All FUAs	Big FUAs	Small FUAs
Mean price per square meter	0.085***	0.073***	0.104*
	(0.024)	(0.019)	(0.056)
Household density	0.296**	0.367***	0.560
	(0.121)	(0.118)	(0.546)
Mean income per adult	−0.004	0.014	−0.030
	(0.011)	(0.013)	(0.019)
Share MF housing	0.777***	0.960***	0.435*
	(0.166)	(0.208)	(0.237)
Homeowners	1.428	2.678*	−0.808
	(1.317)	(1.395)	(2.034)
White collar	−0.010	−0.009	−0.003
	(0.010)	(0.013)	(0.013)
Secondary homes	1.351	−1.248	2.979
	(1.543)	(2.273)	(1.889)
Aged over 60	−0.015	0.020	−0.044
	(0.018)	(0.025)	(0.031)
Observations	58,387	23,929	33,918
Fixed Effects: commune	Yes	Yes	Yes
Fixed Effects: period	Yes	Yes	Yes
Adj. R^2	0.479	0.534	0.341

Table B.1: Determinants of the litigation rate: panel specification.

Note: Commune and period fixed effects, so time-constant covariates (Coastal Law, Mountain Law, distance to the coast) drop out. Weighted by authorized permits, clustered at FUA level. Price, income and density are rescaled to the units used in the cross-section, so coefficients are comparable across tables; this is a change of units only and leaves R^2, N and all t-statistics unchanged. Density here is the census measure, not GHS.

B.2 Unweighted regression

	Anti-construction		
	All FUAs	Big FUAs	Small FUAs
(Intercept)	0.227	0.613*	0.142
	(0.343)	(0.364)	(0.513)
Mean price per square meter	0.101***	0.093***	0.072***
	(0.023)	(0.028)	(0.020)
Household density	0.428***	0.490***	0.038
	(0.050)	(0.028)	(0.092)
Mean income per adult	0.067***	0.098**	0.048**
	(0.025)	(0.038)	(0.024)
Share MF housing	1.192***	1.266***	1.363***
	(0.218)	(0.338)	(0.361)
Homeowners	−0.209	−0.726	−0.559
	(0.294)	(0.490)	(0.459)
Blue collar	0.012**	0.015	0.008
	(0.005)	(0.012)	(0.006)
White collar	0.021	0.016	0.004
	(0.013)	(0.014)	(0.019)
Secondary homes	4.512***	6.234***	3.189***
	(0.544)	(1.233)	(0.697)
Dist. to coast	0.003*	0.004**	−0.001
	(0.002)	(0.002)	(0.005)
Units authorized / stock	−1.731***	−1.668***	−1.672***
	(0.313)	(0.387)	(0.463)
Recreational area	−0.005	−0.008	−0.004*
	(0.003)	(0.006)	(0.002)
Dist. to court	−0.003**	−0.003	0.002
	(0.001)	(0.003)	(0.002)
Observations	22,534	8,864	10,577
Fixed Effects: FUA	Yes	Yes	Yes
Adj. R^2	0.166	0.254	0.070

Table B.2: Determinants of the litigation rate: unweighted.
Note: Same specification as the reduced baseline, without the weighting by authorized permits. Clustered at FUA level.

B.3 Outcome: unit litigation rate

	Anti-construction		
	All FUAs	Big FUAs	Small FUAs
(Intercept)	−4.853**	−3.517*	3.527*
	(2.016)	(1.921)	(2.055)
Mean price per square meter	0.169***	0.149***	0.240**
	(0.053)	(0.053)	(0.098)
Household density	0.243***	0.239**	0.458
	(0.089)	(0.105)	(0.602)
Mean income per adult	0.321**	0.353**	0.212**
	(0.144)	(0.156)	(0.097)
Share MF housing	10.604***	11.770***	6.683***
	(1.133)	(1.205)	(1.399)
Homeowners	3.084***	3.050**	1.601
	(1.034)	(1.341)	(2.980)
Blue collar	0.016	−0.025	0.089*
	(0.031)	(0.043)	(0.049)
White collar	−0.011	−0.054	−0.123
	(0.129)	(0.150)	(0.102)
Secondary homes	8.767***	9.279**	8.313***
	(2.169)	(3.600)	(2.102)
Dist. to coast	0.023**	0.028***	−0.019
	(0.011)	(0.010)	(0.017)
Units authorized / stock	−2.557***	−2.550***	−2.179*
	(0.446)	(0.456)	(1.172)
Recreational area	0.015*	0.021	0.014
	(0.008)	(0.014)	(0.009)
Dist. to court	−0.033***	−0.034***	0.006
	(0.006)	(0.007)	(0.017)
Observations	22,506	8,849	10,570
Fixed Effects: FUA	Yes	Yes	Yes
Adj. R^2	0.477	0.409	0.593

Table B.3: Determinants of the unit litigation rate

Note: This table presents coefficients from the OLS regressions at the municipality level. The dependent variable is the unit litigation rate (share of allowed housing units that face litigation). Columns report results for all FUAs, large FUAs, and small FUAs; columns with "Interc." include an interaction between the homeowners' share and household density. All specifications include FUA fixed effects and the full set of controls described in the text; continuous covariates are median–centered. Standard errors, clustered at the FUA level, are reported in parentheses.

Signif.: : $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

B.4 Alternative variable definitions

	Anti-construction		
	All FUAs	Big FUAs	Small FUAs
(Intercept)	−0.602	−0.403	−0.310
	(0.366)	(0.422)	(0.235)
Household density	0.127***	0.126***	0.091**
	(0.018)	(0.020)	(0.037)
Share MF housing	0.989***	1.002***	1.055***
	(0.166)	(0.215)	(0.223)
Homeowners	−0.795	−2.190***	0.793*
	(0.626)	(0.768)	(0.420)
Blue collar	0.039**	0.060*	0.021**
	(0.019)	(0.033)	(0.008)
White collar	0.097***	0.122***	0.021
	(0.033)	(0.035)	(0.015)
Secondary homes	3.118***	3.687***	2.212***
	(0.509)	(1.009)	(0.439)
Dist. to coast	0.005**	0.006**	0.001
	(0.002)	(0.003)	(0.003)
Units authorized / stock	−1.217***	−1.228***	−0.952***
	(0.294)	(0.335)	(0.185)
Recreational area	0.002	0.003	0.000
	(0.002)	(0.004)	(0.002)
Dist. to court	−0.001	−0.001	0.003*
	(0.003)	(0.004)	(0.002)
Aged over 60	0.026***	0.036**	0.023***
	(0.009)	(0.016)	(0.007)
Median price per square meter	0.107***	0.093***	0.131***
	(0.020)	(0.027)	(0.024)
Median income	0.047	0.115	−0.067**
	(0.058)	(0.083)	(0.031)
Foreigners	0.005	−0.026	0.055**
	(0.021)	(0.022)	(0.023)
Unemployed	−0.015	0.020	−0.039**
	(0.023)	(0.043)	(0.017)
Moutain Law	−0.075	−0.099	−0.109
	(0.107)	(0.177)	(0.138)
Observations	22,514	8,860	10,567
Fixed Effects: FUA	Yes	Yes	Yes
Adj. R^2	0.574	0.587	0.504

Table B.4: Determinants of the litigation rate: alternative definitions.

Note: Mean price per square meter instead of the median, census household density instead of GHS, and the four variables removed elsewhere added back. The squared density term follows the density measure used.

C DETAILED METHODOLOGY FOR DATABASE CONSTRUCTION

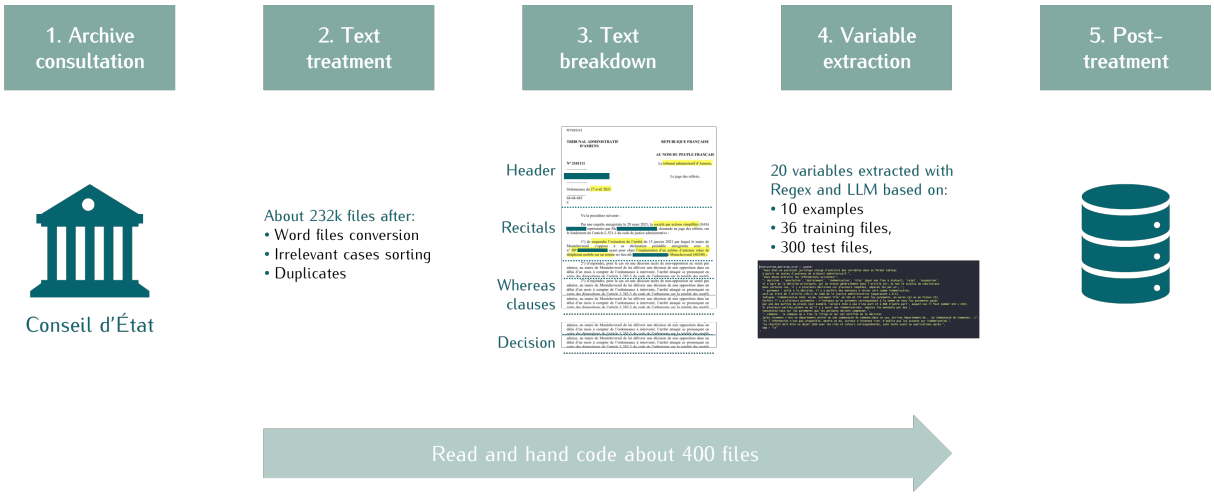


Figure C.1: Main steps in database construction

The following subsections present details of the database construction. Our extraction process relies on custom-developed code combining regular expression patterns and large language model (LLM) processing for variable extraction. The LLM used is `Mistral_small_2503`, which combines the high performance of a model trained on a French corpus, reduced costs and, above all, compliance with data privacy regulations.

C.1 Data collection from legal archives

We construct a novel dataset from French administrative court decisions, obtained from the *Conseil d'État*[11]. We focus on legal decisions concerning urban planning and land use—such as building permits, zoning documents, and preemption procedures—issued by administrative courts (*Tribunaux administratifs*). Targeted keyword searches were conducted on four dates between June and December 2024 as shown in Table C.1. The selection criteria emphasized original court decisions, excluding commentaries or summaries. The search process was time-intensive and iterative, involving trial-and-error to refine keyword relevance and substantial manual review.

Date	Keywords
June 3 and 19, 2024	*permis de construire*
October 8, 2024	*déclaration préalable*, *certificat d'urbanisme*, and cases in the *PCJA68* category (official classification of cases)
November 8, 2024	*autorisation d'urbanisme* and *permis de démolir*
December 20, 2024	*permis d'aménager*, *autorisation d'occupation des sols*, *permis de lotir*, and keywords referring to urban planning documents such as *PLU*, *PLUi*, *plan local d'urbanisme*, and *carte communale*

Table C.1: Archive collection from the *Conseil d'État*

11 Using their repository *Ariane archives* and a less exhaustive but thematically classified server *Ariane*, accessed with the help of the *Centre de recherches et de diffusion juridiques*. It contains historical records of administrative justice in France. The user guide for these searches is available at: https://www.conseil-etat.fr/decisions-de-justice/jurisprudence/rechercher-une-decision-arianeweb

EXHAUSTIVITY CONCERNS There are few sources with precise numbers on administrative court cases that we can evaluate our database against.

Barré et al. (2005) wrote a report on the flows of new cases in the Administrative Courts (ACs) between 1999 and 2005. They mention 5,000 new cases concerning building permits in 1999 and just over 6,000 in 2004. Substantially more than us (2,067 and 3,994 in 2004). We do not know however to what extent the flow of new cases corresponds to the number of court hearings for that period[12] which is our primary data source. It is also likely, from our exchange with civil servants from the *Conseil d'État* (CE) that there might have been larger attrition issues for older cases and that cases deemed trivial were not systematically sent from the local courts to the CE.

For the more recent period, we can also get ballpark estimates of the total number of new cases from the statistical publications of the French ministry of justice. As of 2020, these provide a thematic breakdown of new cases[13], and urban-planning related litigation makes up between 7% of 210,000 new cases received in 2020 to 5% of 257,000 new cases received in 2024 (Chambaz, 2020; Chevalier, 2024). Taking into account that on average about 96% of cases get treated (Conseil d'État, 2023)[14], our dataset would cover about 75% of total cases.

Melot (2015) offer a more direct comparison as they rely on data on court hearings, between 2006 and the end of 2012 for the Courts of Nice and Toulon. They report 8,655 hearings on 'urban planning' over the period compared to 6,493 in our data. From one statistics table (p.9), we compute a total of 4,187 hearings related only to building permits which is closer to our own total of 3,676 (-12%).

It also possible that the data collected from court hearings is somewhat more compact that the source they use[15]. All in all, this suggests that our data may suffer from some attrition but that we cover most of the cases.

C.2 Text treatment

The downloaded documents - mostly in Microsoft Word format - were extracted from bulk ZIP archives and converted into plain text using a combination of automated *Python* scripts and document processing tools. A separate script processed the 1,119 files with alternative formats (.WPD and .RTF), recovering 1,002 usable texts. We built a registry of all the files, each assigned a unique identifier. After standardizing and cleaning the text data, we constructed a unified dataset in `parquet` format. Duplicates - originating from downloads over different keywords - and unreadable files were excluded in post-processing, yielding a clean corpus for analysis. At this stage we have over 230,000 unique cases at the TA level, over 65,000 cases at the CA level and about 9,000 cases at the CE level. The CA and CE cases are reserved for future analysis.

Keyword search has its limits and a significant of cases were only remotely linked to our subject (e.g. applications for a an exemption from local land taxes). In order to isolate relevant legal disputes, we applied a two-step filtering procedure to the introductory claim section of each case:

12 The annual activity reports from the ministry of Justice indicate that the stock is quite stable between 10 and 11 months of average monthly activity between 2014 and 2022 (Camus, 2015) so the flow of new cases should reflect the number of hearings quite closely.

13 Although the composition of these thematic categories is for now not displayed.

14 This is an estimate for all types of litigation, it could be that construction-related litigation is not so well covered given its complexity

15 For example, multiple applications by individuals concerning the same issues are often grouped in one hearing but will be stored separately in the software *Skipper*

- A rule-based classifier using regular expressions filtered out clearly irrelevant decisions (e.g., tax disputes, immigration cases). This first step automatically classifies all observations in the total corpus (69% as relevant, 7% as irrelevant).
- The remaining 24%, for which the recital text does not allow reliable classification using Regex, are considered undefined and are subjected to a second stage of processing by an LLM, which classified them as relevant, irrelevant, or undefined based on inclusion criteria and provided a small justification. Only cases classified as relevant or undefined were retained.

The accuracy of this step is evaluated based on a comparison with 400 files that we have read: 300 files are selected for performance assessment, 36 for training and 10 for examples, the rest is related to irrelevant cases. This residue allows us to validate the performance of this step.

C.3 Text breakdown

On average, the documents are about 6-page long. The text is broken down into sections to reduce costs, and even more so to limit the confusion and hallucinations that an LLM can produce when analyzing a text that is too long (Törnberg, 2023). This is done with Regex using keywords introducing the sections: typically '*Vu la requête*' for statement of facts, '*Considérant*' for court's reasoning and '*DECIDE*' and/or '*Article 1er*' for the decision. An example is given on Figure C.2.

C.4 Variable Extraction

C.4.1 Overview of methodology

We extracted a comprehensive set of variables chosen to capture the legal, procedural, geographic, and substantive dimensions of each case. Table 1 summarizes the extracted variables, their data types, and their origin within the legal document. We employed two main approaches for variable extraction:

- **Regular expressions (R functions):** used for variables that can be reliably extracted from specific keywords or text patterns.
- **Large Language Model (LLM):** applied when relevant information is not consistently located or clearly marked.

Some variables can be extracted from different sections of the text. In this case, we compare the performance of the extraction a posteriori to keep the best one or, in the case of the `operation_municipality`, we combine the different extractions to obtain as comprehensive information as possible. For dates, extraction is mainly done using Regex, and an LLM prompt is used for the few observations for which extraction has not yielded any results.

C.4.2 Prompt engineering

The prompts used for extraction by the LLM contain precise instructions for extracting variables, limiting their length as much as possible. They contain 10 examples (input text and expected output) which allows for both satisfactory performance and cost efficiency (see subsection D.3). They have been iteratively improved by testing the results on a training set of 36 manually coded files as advised in literature (Törnberg, 2023, 2024). Depending on the complexity of the variables, this fine-tuning can take between 5 (as for variables extracted from the outcome, with the second attempt being the best) and 20 steps (as for variables extracted from statement of facts + court's reasoning). In the most complex cases, an LLM is used to improve the prompts. These trials generally reveal trade-offs in accuracy between the

TRIBUNAL ADMINISTRATIF
DE MARSEILLE

N° [redacted]

Mme [redacted]

M. Cousin
Président rapporteur

Mme Markarian
Commissaire du gouvernement

Audience du 20 septembre 2007
Lecture du 4 octobre 2007

RÉPUBLIQUE FRANÇAISE

AU NOM DU PEUPLE FRANÇAIS

Le Tribunal administratif de Marseille

(2ème Chambre)

Vu la requête, enregistrée le 10 décembre 2004, présentée pour Mme [redacted], demeurant 140 [redacted] Marseille (13090), par la SCP [redacted] ; Mme [redacted] demande au tribunal :

- d'annuler la décision en date du 16 juillet 2004 par laquelle le maire de la ville de Marseille a accordé un permis de construire à la SARL [redacted] ;

Considérant que Mme [redacted] demande l'annulation de l'arrêté en date du 16 juillet 2004 par lequel le maire de la ville de Marseille a délivré un permis de construire à la SARL [redacted] afin d'y édifier une maison individuelle d'une surface hors œuvre nette de 77 m² quartier des Goudes ;

D E C I D E :

Article 1er : La requête de Mme [redacted] est rejetée.

Article 2 : Mme [redacted] est condamnée à verser une somme de 1 500 euros à la Sarl [redacted] au titre de l'article L. 761-1 du code de justice administrative.

Article 3: Le présent jugement sera notifié à Mme [redacted], à la ville de Marseille et à la Sarl [redacted].

Figure C.2: Example of the sections selected from a case for analysis
Note: The first image shows the header and statement of facts. The second shows the 'court's reasoning' (particularly short here). The third shows the outcome.

different variables extracted simultaneously. The final prompts are presented below and compiled in a prompt codebook.

Script 1: Statement of facts extraction

```
"Vous êtes un assistant juridique chargé d'extraire des variables dans un format tableau
    à partir de textes d'audience de tribunal administratif.

Vous devez extraire les variables suivantes:
- adresse_requ : l'adresse des requérants

- commune_requ : la commune de résidence des requérants. Si multiples occurences de la mê
    me commune, ne mettre chacune qu'une seule fois.

- code_commune_requ : le code INSEE de la commune de résidence des requérants. Si
    multiples occurences du même commune, ne mettre chacun qu'une seule fois.

- commune_litige : la commune où a lieu le litige. (En cas d'hésitation, davantage que l'
    entité visée c'est le lieu du litige qui compte). Rarement c'est un département ou un
     EPCI entier. Dans ce cas le préciser, par exemple en mettant 'Département de' ou le
    type d'EPCI.

- objet : l'objet du litige. Un ou plusieurs (séparés par ';') parmi : 'PC' pour permis
    de construire',
  'DP' pour déclaration préalable ou déclaration de travaux, 'PA' pour permis d'aménager,
  'CU' pour certificat d'urbanisme, 'PD' pour permis de démolir, 'PLU' ou 'PLUi' pour
      plan local d'urbanisme (i si intercommunal),
  'CC' pour Carte communale, 'PSMV' pour plan de sauvegarde et de mise en valeur,
  'POS' pour plan d'occupation des sols, 'Préemption' si ce droit est exercée pour pré
      empter,
  'Indemnisation' quand il s'agit d'une compensation financière,
  'Autres opérations' pour d'autres types de travaux ou la préemption,
  'Autres DA' pour d'autres types de documents d'aménagement

- type_litige : positif si le litige concerne un arrêté qui a été accordé/approuvé (un
    permis de construire, un certificat, un document d'urbanisme, une absence d'
    opposition à DP ou autre,) par le responsable (souvent le maire), négatif si le
    litige concerne un refus

- type_requerant : un ou plusieurs (séparés par ';') parmi: Association, Particulier, Pré
    fet, SCI (si SCCV mets SCI), Société, Commune, ou Autre Etat. Si multiples occurences
     du même type de requerant, ne mettre chacun qu'une seule fois.

- avocat : lorsque les plaignants sont représentés par un avocat.

Si l'information n'est pas disponible, mettre un NC, n'inventez rien.
Le résultat doit être un objet JSON avec les clés et valeurs correspondantes, sans texte
    avant ou explications après."
```

Script 2: Statement of facts + court's reasoning (i) extraction

```
"Vous êtes un assistant juridique chargé d'extraire des informations depuis des textes d'
    audiences de tribunal administratif.

Votre mission :
 - Extraire les variables demandées sous forme d'un objet JSON.
 - Respecter strictement les définitions et les règles ci-dessous.
 - Ne jamais créer d'information non présente dans le texte.

Variables à extraire :
```

```
1. type_litige :
   - 'positif' si le litige porte sur un arrêté accordé ou approuvé (permis de construire
     , certificat, document d'urbanisme, une absence d'opposition à DP, etc.) par le
     responsable (souvent le maire)
   - 'négatif' si le litige porte sur un refus.

2. adresse_operation :
   - Adresse ou lieu-dit de l'opération (sans la commune).
   - Ne pas indiquer le numéro de parcelle/terrain ou l'adresse des demandeurs.
   - Mettre NA si le litige ne correspond pas à une opération ciblée

3. commune_operation :
   - Commune du litige.
   - Si département ou EPCI, préciser par ex : 'Département de ...' ou 'Communauté de
     communes ...'.

4. classement_parcelle :
   - Booléen :
   - TRUE si classement parcelle en non constructible dans le cadre de la révision d'un
     document d'urbanisme,
   - FALSE sinon,
   - Mettre NA si objet n'est pas document d'urbanisme (PLU, PLUi, PSMV, POS, Autres DA)
     ou si le litige concerne une division de parcelle

Règles générales d'extraction :
 - Si l'information est absente, indiquer 'NC'.
 - Ne jamais inventer d'information ou interpréter au-delà du texte.
Format de sortie attendu :
 - Un objet JSON brut, sans texte supplémentaire, sans explication."
```

Script 3: Statement of facts + court's reasoning (ii) extraction

```
"Vous êtes un assistant juridique chargé d'extraire des variables dans un format tableau
    à partir de textes d'audience de tribunal administratif.
Vous devez extraire les variables suivantes:

logement :
   - TRUE si le cas concerne des logements (y compris collectifs ou en résidence). FALSE
     si c'est autre chose. NC si l'information n'est pas disponible.

nb_logement :
   - Si logement=TRUE, le nombre total de logements construits (faire la somme).
   - Si le cas ne concerne pas de logements mettre NA
   - mettre 'NC' si l'information n'est pas disponible (par exemple des logements
     collectifs sans détail sur le nombre).

logement_social :
   - TRUE si le cas concerne des logements sociaux (HLM, etc. mais pas EHPAD). En général
      c'est écrit explicitement.

nb_logement_social :
   - Si logement_social=TRUE, le nombre total de logements construits (faire la somme).

type_construction :
   - Modalité parmi la liste suivante uniquement :
     - 'Logement individuel construction' qui comprend les dépendances
     - 'Logement individuel extension rénovation' : amélioration de l'existant (véranda
       par exemple)
     - 'Logement individuel groupé' : lotissements de maisons
```

```
    - 'Logement collectif' : immeuble d'appartements hors résidences
    - 'Logement en résidence' : exemple résidence étudiante, EHPAD
    - 'Annexe habitat' : constructions séparées du logement. Par exemple garage, piscine
      , cloture, cabane
    - 'Eolien', 'Méthaniseur', 'PV', 'Réseau / raccordement', 'Antennes' (relais télé
      phonique),
    - 'Transport' : voiries, rail, station-service,
    - 'Agricole' : uniquement exploitation agricole ou forestière
    - 'Artisanat', 'Bureaux',
    - 'Services publics' : enseignement, action sociale, santé, culture, loisirs, sport,
      bâtiments religieux,
    - 'Commerces', 'Entrepôts' : site logistique..., 'Industrie',
    - 'Services publics',
    - 'Tourisme' : hôtels, camping,
    - 'Division parcelle' : quand on divise explicitement les parcelles,
    - 'Autres infrastructures'
    - 'NA' uniquement si ne porte pas sur un projet.

Règles générales d'extraction :
Si l'information n'est pas disponible, mettre un NC, n'inventez rien.
Le résultat doit être un objet JSON avec les clés et valeurs correspondantes, sans texte
    avant ou explications après."
```

Script 4: Outcome extraction

```
"Vous êtes un assistant juridique chargé d'extraire des variables dans un format tableau
    à partir de textes d'audience de tribunal administratif.
Vous devez extraire les informations suivantes :

- decision : 'annulation', 'désistement', 'indémnisation', 'nlas' (pour non lieu à
    statuer), 'rejet', 'suspension'.
  Il s'agit de la décision principale, qui se trouve généralement dans l'article 1er, et
      non le surplus de conclusions.
  Dans certains cas, il y a plusieurs décisions car plusieurs requêtes, séparez les par
      un ;.

- paiement : suite à la décision, il y a parfois des montants à verser soit comme indé
    mnisation, soit au titre de l'article L761-1 du code de la justice administrative (
    auparavant L.8-1).
  Indiquez 'indémnisation XXXe' et/ou 'paiement YYYe' où XXX et YYY sont les paiements,
      en euros (e) ou en Francs (F).
  Parfois il y a plusieurs paiements : n'indiquez qu'un paiement correspondant à la somme
       de tous les paiements payés par une des parties du procès (par exemple 'versera
      XXXe à AAA d'une part et à BBB d'autre part', auquel cas il faut sommer XXX + XXX).
  Si plusieurs parties paient ou qu'il y a aussi des indemnisations, séparez les montants
       par des ;. Concentrez-vous sur les paiements que les perdants doivent compenser.

- commune : la commune où a lieu le litige et qui est notifiée de la décision (plus
    rarement c'est un département entier ou une communauté de commune, dans ce cas, é
    crivez Département de... Ou Communauté de Communes...)

Si l'information n'est pas disponible, mettre un NA, surtout n'inventez rien. N'oublie
    pas les accents sur indémnisation.

Le résultat doit être un objet JSON avec les clés et valeurs correspondantes, sans texte
    avant ou explications après."
```

C.4.3 Description of variables

The categories of the factorial variables are described below:

- **`court` is one of the 42 local courts:** Amiens, Bastia, Besançon, Bordeaux, Caen, Cergy-Pontoise, Châlons-en-Champagne, Clermont-Ferrand, Dijon, Grenoble, Guadeloupe, Guyane, La Réunion, Lille, Limoges, Lyon, Marseille, Martinique, Mayotte, Melun, Montpellier, Montreuil, Nancy, Nantes, Nice, Nîmes, Nouvelle-Calédonie, Orléans, Paris, Pau, Poitiers, Polynésie Française, Rennes, Rouen, Saint-Barthélemy, Saint-Martin, Saint-Pierre-et-Miquelon, Strasbourg, Toulon, Toulouse, Versailles, Wallis et Futuna;
- **`administrative_act`:** categories are described in Table C.2;
- **`initial_outcome`:** either one of these values
  - *positif* when the project or urban planning document has been accepted (typically someone contesting their neighbor's building permit approval)
  - *négatif* when the project or urban planning document has been rejected (typically someone contesting their own building permit rejection)
- **`litigant_type`:** categories are described in Table C.3;
- **`construction_type`:** categories are described in Table C.4;
- **`outcome`:** categories are described in Table C.5.

C.4.4 Descriptive statistics on extracted variables

Table C.6 and Table C.7 present descriptive statistics on numerical and categorical variables respectively. Figure C.3 shows the number of cases either by date of application or by date of hearing. We observe a sharp increase in the number of cases until the mid-2000s, most likely due to the gradual digitization of archives, followed by a slowdown in the increase, with the level then approaching the trends observed for the annual increase in litigation, and then stabilizing from the early 2010s onwards. Therefore, and by comparison with existing literature (subsection C.1), we retain post-2005 data for analysis. Furthermore, as the data was collected during 2024 (on all past hearings), the year is incomplete and we retain cases with a hearing date in 2023 or earlier.

Category	Code	French label	English translation
Project specific	PC	Permis de construire	Building permit
	DP	Déclaration préalable	Prior development notice
	PA	Permis d'aménager / permis de lotir	Planning permission /subdivision permit
	CU	Certificat d'urbanisme	Urban planning certificate
	PD	Permis de démolir	Demolition permit
	-	Indemnisation	Compensation for lost earnings on rejected projects, overrun works
	-	Autres opérations	Other operations: unauthorized works, preemption, land clearing, public utility works
Planning document	PLU	PLU	Local urban plan
	PLUi	PLUi	Inter-municipal local urban plan
	CC	Carte communale	Municipal map
	PSMV	Plan sauvegarde et mise en valeur (vaut PLU)	Heritage preservation and enhancement plan (equivalent to PLU)
	POS	Plan d'occupation des sols	Land use plan
	-	Autres DA	Other planning documents, development schemes

Table C.2: Categories of the variable `administrative_act`

French label	English translation
Association	Association
Commune	Municipality
Particulier	Individual
Préfet	Appointed representative of the French central government at the departmental level, responsible for state administration and public order
SCI	A French civil law entity for property ownership and management, excluded from commercial activities
Société	Company
Autres	Others

Table C.3: Categories of the variable `litigant_type`

Category	French label	English translation	Note
Housing	Logement individuel construction	Individual housing construction	Including detached house in the garden
	Logement individuel extension rénovation	Individual housing extension/renovation	Housing quality improvement
	Logement individuel groupé	Grouped individual housing	Entire subdivision
	Logement collectif	Collective housing	
	Immeuble	Building	
	Logement en résidence	Residential housing	Nursing home, student housing...
	Annexe habitat	Housing annex	Garage, swimming pool, fence, shed, canopy, boiler room...
Infrastructure	Éolien	Wind energy	
	Méthaniseur	Biogas plant	
	PV	Photovoltaic	
	Réseau / raccordement	Network / connection	
	Antennes	Telephone relay	
	Transport	Transport	Roads, rail, service station
	Autres infrastructures	Other infrastructure	Artificial water reservoir for agriculture, fences (fields)
Economic activity	Agricole	Agricultural	Buildings or equipment excluding biogas plants, including forestry
	Artisanat	Crafts	
	Bureaux	Offices	
	Commerces	Retail	Including restaurants, multiple parking spaces...
	Entrepôts	Warehouses	Including logistics platforms
	Industrie	Industry	
	Services publics	Public services	Education, social action, health, culture, leisure, religious buildings
	Tourisme	Tourism	Hotel accommodation, camping...
Zoning	Division parcelle	Plot division	

Table C.4: Categories of the variable `construction_type`

French label	English translation
Annulation	Cancellation
Désistement	Withdrawal
Rejet	Rejection
Indemnisation	Compensation
Non lieu à statuer	No grounds for ruling or stay of proceedings
Suspension	Suspension

Table C.5: Categories of the variable `decision`

Variable	Min	Med	Mean	Max	# NA
date_application	1995-01-02	2012-08-16	2012-02-14	2024-12-21	0 (0%)
date_hearing	1995-01-04	2013-11-29	2013-07-05	2024-12-19	0 (0%)
housing_number	0	1	10	160	81 (0.1%)
social_housing_number	0	6	19	247	25 (0.4%)
compensation	0	0	1,541	39,267,281	0 (0%)
irreducible_costs	0	0	454	3,000	0 (0%)

Table C.6: Summary statistics and distribution of extracted variables - Dates and numerical variables
Note: Computed on the full database. "# NA" is the number (share) of missing values. For `housing_number` and `social_housing_number`, statistics and missing values are computed only on the cases where `housing` (respectively `social_housing`) is true, i.e. 45.1% (3.1%) of the database; both variables are winsorized at the 1st and 99th percentiles. `compensation` and `irreducible_costs` are set to 0 when no amount is mentioned, so they have no missing value; `irreducible_costs` is winsorized at the 99th percentile. Amounts are in 2024 euros (francs are converted with a factor of 0.23).

Variable	# NA	Most frequent values
court	11 (0%)	Grenoble (17,494, 7.8%); Marseille (14,772, 6.6%); Rennes (13,977, 6.2%); Lyon (11,624, 5.2%); Versailles (11,075, 4.9%)
operation_municipality	906 (0.4%)	Paris (2.1%); Marseille (1.3%); Toulouse (0.5%); Nice (0.4%); Aix-en-Provence (0.4%)
operation_address	103,452 (46.2%)	
administrative_act	114 (0.1%)	Building permit (107,366, 48%); Prior declaration of works (26,973, 12.1%)
initial_outcome	7,070 (3.2%)	Granted (146,056, 67.4%); Denied (70,786, 32.6%)
plot_classification	0 (0%)	FALSE (201,544, 90%)
applicant_type	119 (0.1%)	Individual (137,983, 61.7%); Company (24,401, 10.9%)
applicant_municipality	110,039 (49.1%)	Paris (4.5%); Marseille (1.7%); Nice (0.6%); Bordeaux (0.4%); Nantes (0.4%)
applicant_address	112,610 (50.3%)	
lawyer	436 (0.2%)	TRUE (154,045, 68.9%); FALSE (69,370, 31%)
housing	44,227 (19.8%)	TRUE (100,886, 56.1%); FALSE (78,793, 43.8%)
social_housing	51,953 (23.2%)	FALSE (165,015, 96%)
construction_type	46,253 (20.7%)	Individual housing: new construction (31,117, 17.5%); Collective housing (18,648, 10.5%)
outcome	3,300 (1.5%)	Rejection (113,123, 51.3%); Cancellation (39,065, 17.7%); Withdrawal (35,240, 16%)

Table C.7: Summary statistics and distribution of extracted variables - Factor and string variables
Note: Computed on the full database. "# NA" is the number (share) of missing values. "Most frequent values" lists the categories that are among the five most frequent and represent at least 10% of the non-missing observations, with their count and share of the non-missing observations (so only one category is shown for `plot_classification` and `social_housing`); for municipalities only the share is shown, and addresses are not summarized. Petitioner's municipality and address are not reported in decisions after 2015 and are set to missing. Category labels are translated from French.

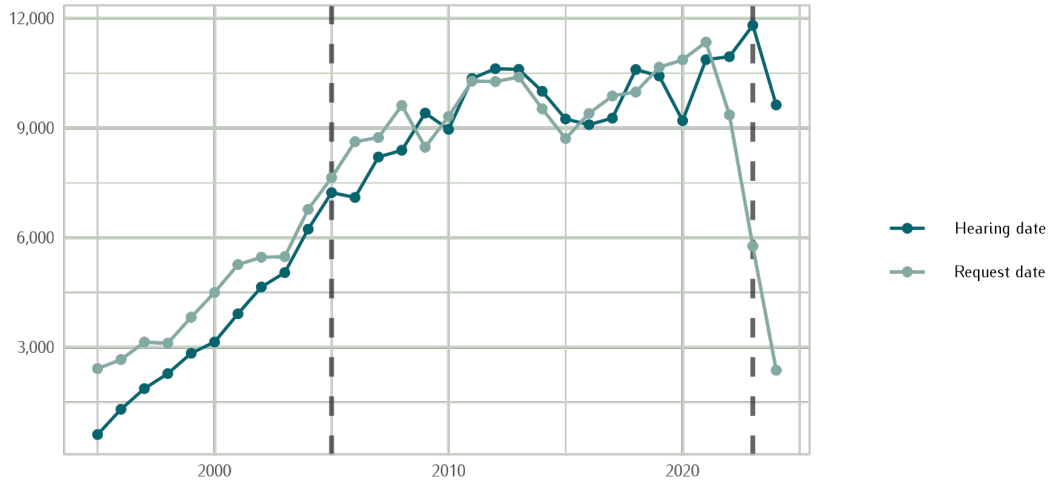


Figure C.3: Litigation cases per year
Note: The number of cases is plotted either by date of application or by date of hearing, with hearings taking place on average two years after the application.

C.5 Post treatment

C.5.1 Matching municipality names to official codes

We develop a procedure to match the municipality names mentioned in court decisions to official French municipal codes (code INSEE). The first step is to get a clean municipality name. We use the municipality variable extracted from the statement of facts as our baseline as it is better spelled[16] in most cases. When the statement of facts did not allow to get the name of the municipality concerned by the litigation[17], we use the municipality names extracted from the others parts of the text[18]. The municipality variables from the different sources are close with more than 90% exact match and more than 95% for a fuzzy distance of two.
We exclude the following observations from the matching process:

- Missing municipality: #obs = 910
- higher-level administrative unit then municipality: #obs = 4670
- Multiple municipalities in one case: #obs = 1,173 (we address these at the end)
- Date of application or date of audience outside of the 1990-2024 period: #obs =1, 006

The matching procedure then involves four steps:

1. **Court-Department Mapping:** We construct a correspondence table linking each Administrative Court to the *Départements* under its jurisdiction. This mapping accounts for jurisdictional changes over the period (new courts opened in *Île-de-France* and *Provence-Alpes-Côte d'Azur* (PACA) for instance).

16 (less missing articles, accents, hyphens etc.)

17 This can happen when a higher-tier local government gets litigated regarding planning documents or large-scale litigation for instance

18 one rare exception is when the statement of facts text has a different municipality name than the other two sources, which match; in such cases, we keep the matching ones (see R script for details).

2. **Exact Matching:**[19] For each case, we attempted to match the extracted municipality name to the list of officially recognized communes for the corresponding year and *Départements* under the court's jurisdiction. The temporal dimension accounts for historical changes in commune names or boundaries.
   To uniquely identify duplicates resulting from within-jurisdiction homonyms (approximately 2,000) we use the extracted address when available and usable (about 30% of the time). Unresolved duplicated cases get a missing value for the municipality code.

3. **Fuzzy Matching:** If exact matching failed, we applied a fuzzy string matching algorithm to identify likely candidates among valid commune names. We applied a modified Optimal String Alignment (OSA) procedure: it accepts an initial OSA distance of 3, which we adjust by discounting changes involving articles, hyphens, or apostrophes, and retain matches with a modified distance ≤ 2.
   We then deal the duplicated cases as with exact matching and retain only the most confident matches.

4. **Manual corrections:** we perform a few manual corrections. Typically this involves cases manually correcting cases where the judicial administration refers to a named neighborhood instead of the municipality or a quasi-official nickname (something quite common in coastal resort towns)[20].

Multiple municipalities:
Cases that involve larger infrastructure such as energy projects (Wind energy, methane installations etc.) typically involve more than one municipality. There are just over 1,200 such cases. We flag these observations with a boolean variable so that they may be used separately from the others by the user. Then we adopt a 'municipality point of view' where we duplicate these cases once per municipality involved. Thus the data will reflect that for a given municipality there has been a land-use related litigation on its territory. We only rely on the exact matching part of the procedure for these observations and match 85% of all municipalities involved (95% of the cases are matched to at least one municipality).

We successfully match **216,668** unique cases to a municipality, i.e. 96,4% of our dataset with clean municipality names and 93,2% of the total dataset.

C.5.2 Geocoding the plaintiffs' addresses

For the 2000-2012 period we were able to recover a usable address for the plaintiff for about 87% to 92% of cases depending on the year[21]. To assign geographic coordinates to this address-level data, we employed an automated geocoding procedure based on the official French national address API[22].
The geocoding pipeline consisted of the following steps.
Each address is first standardized by concatenating available street and number information with the municipality field to form a geocoding query.
These are then submitted to the French national address API. When a result is returned, the response includes a standardized address label, latitude and longitude coordinates, and a confidence score indicating the quality of the match (from 0,3 to 1).

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19 We thank the author of the R package `COGugaison` for making this task much simpler.

20 These modifications are all visible in the R codes provided

21 The years in the late 1990s could also have been used but they suffer from increasing attrition of cases as we go back in time. The share of missing addresses increases rapidly after 2012 (33% in 2013, 65% in 2014).

22 https://api-adresse.data.gouv.fr

Although the procedure is the same, we process cases of single and multiple-addresses (several address per litigation case, separated by a ';') separately. The user must note that multiple cases result in duplicated case IDs.

There are just over 100,000 cases between 2000 and 2012, of which about 91,6K have an address for the plaintiff. The quality of the addresses enables us to geocode just short of 98% of the cases with a high degree of certainty as shown by subsubsection C.5.2.

	Single	**Multiple**	**Total**
# unique cases	75 510	16 162	91 672
Matched	73 503	16 077	89 580
p25 score	64.8	62.4	-
p50 score	93.8	93.2	-
p75 score	96.3	96.2	-

The user must decide what the appropriate level of quality is for safe use. Taking 55% as a threshold removes the bottom-quintile and yields good result upon manual verification.

C.5.3 Other data sources

To enrich the constructed database, several additional data sources were integrated across multiple dimensions. Population and socio-demographic characteristics of municipalities including socio-professional categories, employment density, unemployment rates, income levels, foreign-born population shares, counts of primary and secondary residences and tourism capacity, were obtain from INSEE's population census. To calculate a measure of population density that reflects the number of people in the urbanized environment, we divide the number of households in the municipality by the built surface obtained from the Global Human Settlement Layer data [23]. Environmental amenities were derived from multiple sources: GIS layers from the National Inventory of Natural Heritage (INPN), slope data from IGN's BD ALTI 75m, coastal distance computed using the `cpprouting` R package, altitude data from the municipalities dataset on data.gouv.fr [24], national and regional parks information from the *Observatoire des territoires*, recreational spaces (green areas and water bodies) from CORINE Land Cover[25], and coastal law municipalities[26] and mountain law municipalities[27] from data.gouv.fr. Housing data incorporated average price per square meter from DV3F transactions[28]. Building permits (granted or denied) and associated housing units from Sitadel[29]. Finally the harmonization of municipal codes was made much less painful by the admirable R package `COGugaison`[30] for which we are thankful.

23 https://human-settlement.emergency.copernicus.eu/

24 https://www.data.gouv.fr/datasets/communes-et-villes-de-france-en-csv-excel-json-parquet-et-feather/reuses_and_data services

25 https://www.statistiques.developpement-durable.gouv.fr/corine-land-cover-0

26 https://www.data.gouv.fr/datasets/communes-de-la-loi-littoral-au-code-officiel-geographique-cog-2020-2022/

27 https://www.data.gouv.fr/datasets/communes-de-la-loi-montagne-au-code-officiel-geographique-cog-2020-2022/

28 https://datafoncier.cerema.fr/dv3f

29 https://www.data.gouv.fr/datasets/base-des-permis-de-construire-et-autres-autorisations-durbanisme-sitadel/

30 https://github.com/antuki/COGugaison

C.6 Data cleaning

C.6.1 Handling duplicate cases

To minimize issues of double counting, we remove litigation cases that may correspond to the same building permit[31], something more common for large construction projects. Starting from 40,911 cases against granted housing permits filed in mainland France between 2005 and 2022, we apply six successive rules, from the most to the least reliable. A case grouped at one step is not considered in the following ones.

Step 1: We group cases matched to the same permit in the Sitadel database, using exact or controlled matches on the permit number checked for timing consistency (6,684 cases, 5,840 permits).

Step 2: Among the remaining cases, we group those citing the same complete permit number in the same municipality, as extracted from the text of the decisions (117 cases, 45 permits).

Step 3: We geocode litigation addresses with the national address database (BAN), keeping matches with a score of at least 0.80 located in the litigation municipality. We group cases at the same address number reporting the same positive number of units (430 cases, 204 permits).

Step 4: We group cases located in the same street that report the same positive number of units and construction type and were filed at most three years apart, requiring every pair within a group to meet these conditions (164 cases, 75 permits).

Step 5: We extract cadastral parcel references from the decisions and manually review the 890 pairs of cases citing the same parcel and sharing the same address, number of units or construction type. 669 pairs refer to the same permit, which yields 980 cases and 453 permits.

Step 6: We review in the same way the 4,215 pairs of cases with the same address, number of units and construction type, filed at most three years apart. 2,616 pairs refer to the same permit, which yields 3,320 cases and 1,495 permits.

In steps 5 and 6, groups containing a pair judged to involve different permits are split into subsets in which all pairs refer to the same permit. Each group is then collapsed into a single observation, using the most frequent non-missing value of each variable. This reduces the sample from 40,911 cases to 37,328 litigated permits.

We then winsorize the number of litigated housing units at the 99.9^{th} percentile (358 units). Cases against collective housing reporting fewer than two units or none (25% of collective-housing cases) are assigned the mean size of authorized collective permits in their municipality, or in their department when the municipality has seven or fewer such permits. Other cases with a missing or zero number of units are counted as one unit. Our final sample includes 37,328 litigated housing permits in mainland France, representing 1.3% of all granted permits over the period. Multi-family (MF) housing cases represent 36% of the sample. The total number of housing units in litigated permits is 464,138 without imputation and 550,767 without imputation, respectively 6,4 and 7.6% of authorized housing units.

31 It is frequent for large projects to be attacked by several individuals, an NGO etc.

D PERFORMANCE ASSESSMENT

The following subsection provides additional details on the LLM's performance assessment. The evaluation process is tailored to the variables and use of the database.

D.1 Metrics for performance assessment

We use standard metrics for evaluating the performance of an LLM (more generally, algorithmic coding evaluated against human coding) on all variables (including those extracted with Regex functions) (Leek et al., 2024; Pelaez et al., 2024; Törnberg, 2023, 2024).
Accuracy is the proportion of correct predictions out of total predictions. The Mean Absolute Error (MAE) is the average of absolute differences between predicted and actual values. The Root Mean Square Error (RMSE) is the square root of the average squared differences between predicted and actual values. Precision is the proportion of positive predictions that were actually correct. Recall is the proportion of actual positives that were correctly identified. F1 scores evaluate the balance between precision and recall. F1 Micro gives equal weight to each sample, which favors frequent classes. F1 Macro gives equal weight to each class, which is preferable when classes are unbalanced. Table D.1 presents performance results across all variables.

For all variables

$$\text{Accuracy} = \frac{\sum_{i=1}^{n} \mathbb{1}(y_i = \hat{y}_i)}{n}$$

For numeric variables

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^{n} |y_i - \hat{y}_i|$$

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^{n} (y_i - \hat{y}_i)^2}$$

For factor/logical variables

$$\text{Precision}_k = \frac{\text{TP}_k}{\text{TP}_k + \text{FP}_k}$$

$$\text{Recall}_k = \frac{\text{TP}_k}{\text{TP}_k + \text{FN}_k}$$

$$\text{F1}_k = \frac{2 \cdot \text{Precision}_k \cdot \text{Recall}_k}{\text{Precision}_k + \text{Recall}_k}$$

$$\text{F1}_{\text{micro}} = \frac{2 \cdot \sum_k \text{TP}_k}{2 \cdot \sum_k \text{TP}_k + \sum_k (\text{FP}_k + \text{FN}_k)}$$

$$\text{F1}_{\text{macro}} = \frac{1}{K} \sum_{k=1}^{K} \text{F1}_k$$

where TP is 'True positive', FP is 'False positive', FN is 'False negative', k is one category of the variable.

Computations are adapted to the specificities of the variables. For `date_application` and `date_hearing`, the final scores are a combination of the performance of the regex process and the LLM on the remaining cases. MAE and RMSE are expressed in: days for dates, number of dwellings for `housing_number` and `social_housing_number` and euros 2024 for `compensation` and `irreducible_costs`. For the dwellings, the metric is calculated only when the `housing` dummy (`social_housing` dummy, respectively) is TRUE, so as not to double count the errors of the dummies. For `operation_municipality`, accuracy is slightly underestimated: the variable is extracted from the statement of facts, the court's reasoning and the outcome. The accuracy of the best of these three results (statement of facts + court's reasoning) is given here; the other variables used on missing values increase the final score (see subsection C.5 of the appendix). For string variables, capitalization is not taken into account for evaluation.

Variable	Accuracy	MAE	RMSE	Precision	Recall	F1 micro	F1 macro
`date_application`	99.3%	0.07	1.27	-	-	-	-
`date_hearing`	98.3%	0.14	1.82	-	-	-	-
`court`	98.7%	-	-	100.0%	98.7%	99.3%	99.1%
`operation_municipality`	96.3%	-	-	-	-	-	-
`operation_address`	82.7%	-	-	-	-	-	-
`administrative_act`	93.0%	-	-	94.7%	94.2%	94.4%	85.4%
`initial_outcome`	94.3%	-	-	95.9%	94.3%	95.1%	95.0%
`plot_classification`	93.7%	-	-	93.7%	93.7%	93.7%	67.7%
`applicant_type`	95.7%	-	-	97.3%	98.7%	98.0%	93.2%
`applicant_municipality`	98.3%	-	-	-	-	-	-
`applicant_address`	97.7%	-	-	-	-	-	-
`lawyer`	97.3%	-	-	97.7%	97.3%	97.5%	96.8%
`housing`	94.0%	-	-	94.0%	94.0%	94.0%	94.0%
`housing_number`	87.4%	0.83	5.63	-	-	-	-
`social_housing`	99.3%	-	-	99.3%	99.3%	99.3%	92.7%
`social_housing_number`	100.0%	0.00	0.00	-	-	-	-
`construction_type`	83.3%	-	-	84.5%	87.9%	86.2%	83.0%
`outcome`	95.7%	-	-	97.8%	97.6%	97.7%	84.5%
`compensation`	98.7%	14.73	130.34	-	-	-	-
`irreducible_costs`	97.0%	57.67	438.75	-	-	-	-

Table D.1: Assessment of performance on 300 human coded files

Notes: 1. For `operation_municipality`, accuracy is slightly underestimated: the variable is extracted from the statement of facts, the court's reasoning and the outcome. The accuracy of the best of these three results (statement of facts + court's reasoning) is given here; the other variables used on missing values increase the final score (see subsection C.5 of the appendix). For `date_application` and `date_hearing`, the final scores are a combination of the performance of the regex process and the LLM on the remaining cases. **2.** MAE and RMSE are expressed in: days for dates (compared to a range of approximately 30 years), number of dwellings for `housing_number` (mean of the variable: 11.3) and `social_housing_number` (mean of the variable: 15.4) and euros 2024 for `compensation` (mean: €1,041) and `irreducible_costs` (mean: €541). **3.** For the dwellings, the metrics are calculated only when the `housing` dummy (`social_housing` dummy, respectively) is TRUE. This is the case for 147 observations (7 observations, respectively, for which the prediction turns out to have no error). **4.** For factor variables, scores are calculated excluding the 'Unknown' class from the calculation as the aim is to accurately predict the other classes. We also separate composite categories (e.g. 'PC; CU' for `administrative_act`) into n categories weighted by $1/n$ (here 'PC' with a weight of 0.5 and 'CU' with a weight of 0.5) in accordance with database usage. Alternative metrics (Table D.2) show that this choice does not compromise the reliability of the database.

D.2 Alternative performance scores for factor variables

When calculating scores for factor variables, two choices are made in the calculation, related to the use of the database. Here, we evaluate the impact of these choices on performance metrics and show that the results remain reliable.

- Scores are calculated excluding the 'Unknown' (*'Non connu'*) class from the calculation as the aim is to accurately predict the other classes.

- We also separate composite categories (e.g. 'PC; CU' for `administrative_act`) into n categories weighted by $1/n$ (here 'PC' with a weight of 0.5 and 'CU' with a weight of 0.5) in accordance with database usage. This allows to assess if the answer is partially correct.

Alternative metrics in Table D.2 show that this choice does not compromise the reliability of the database.

Variable	Accuracy	With NC - Aggregated		No NC - Aggregated		With NC - Weighted		No NC - Weighted	
		F1 micro	F1 macro	F1 micro	F1 macro	F1 micro	F1 macro	F1 micro	F1 macro
`court`	98.7%	98.7%	99.1%	99.3%	99.1%	98.7%	99.1%	99.3%	99.1%
`administrative_act`	93.0%	93.0%	82.3%	93.0%	82.3%	94.4%	85.4%	94.4%	85.4%
`initial_outcome`	94.3%	94.3%	95.0%	95.1%	95.0%	94.3%	95.0%	95.1%	95.0%
`plot_classification`	93.7%	93.7%	67.7%	93.7%	67.7%	93.7%	67.7%	93.7%	67.7%
`applicant_type`	95.7%	95.7%	95.6%	95.7%	95.6%	98.0%	93.2%	98.0%	93.2%
`lawyer`	97.3%	97.3%	96.8%	97.5%	96.8%	97.3%	96.8%	97.5%	96.8%
`housing`	94.0%	94.0%	94.0%	94.0%	94.0%	94.0%	94.0%	94.0%	94.0%
`social_housing`	99.3%	99.3%	92.7%	99.3%	92.7%	99.3%	92.7%	99.3%	92.7%
`construction_type`	83.3%	83.3%	85.2%	84.7%	85.5%	84.6%	82.7%	86.2%	83.0%
`outcome`	95.7%	95.7%	88.5%	96.5%	88.5%	96.9%	84.5%	97.7%	84.5%

Table D.2: Sensitivy tests on F1 scores computed on categorical variables
Note: 'NC' stands for 'Non Connu' and is considered either as a class or removed from computations. Aggregate metrics are calculated with composite classes ('PC;CU' is one class, 'PC' and 'CU' are two other classes), weighted metrics are calculated with single element classes, the sum of the elements of an observation being 1 ('PC; CU' contributes 1/2 to 'PC' and 1/2 to 'CU').

D.3 Examples of inaccurate coding

To illustrate the type of errors made during data extraction, 5 errors were randomly selected from the incorrectly coded variables. The results are shown in Table D.3.

Case	Variable	Hand coded	LLM/Regex coded
1	`administrative_act`	DP	Autres opérations
2	`operation_municipality`	NC	Département de la Haute-Saône
3	`date_hearing`	2000-06-20	2000-07-04
4	`irreducible_costs`	1000	0
5	`housing`	FALSE	TRUE

Table D.3: Random sample of 5 errors among mismatch between true and predicted value
Note: 'DP' stand for *Déclaration préalable*, a type of building permit required for small operations. *Autres opérations* means 'Other operations'.

The errors in each case can be explained as follows:

1. The text provided for data extraction is insufficient to determine that it is a *Déclaration préalable* since the information is found before the extract. This is due to the fact that the functions used to split the text are imperfect and cannot handle the rare cases where the tags indicating a new section in the text are missing.
   - **Text provided for evaluation by LLM:**
   Vu les autres pièces du dossier.
   Vu :
   - le code de l'urbanisme ;
   - le code de justice administrative.
   La présidente du tribunal a désigné Mme [redacted] *pour statuer sur les demandes de référé. Les parties ont été régulièrement averties du jour de l'audience. Après avoir, au cours de l'audience publique du 6 novembre 2019, présenté son rapport et entendu les observations de Me* [redacted] *pour la société* [redacted] *et de Me* [redacted] *pour la commune de Tauxigny-Saint-Bauld, qui ont repris les éléments exposés par écrit.*
   Après avoir, à l'issue de l'audience publique, prononcé la clôture de l'instruction. Considérant ce qui suit :

Sur les conclusions à fin de suspension :
1. La société [redacted] *indique être un opérateur de communications électroniques au sens du code des postes et communications électroniques*

- **Actual statement of facts:**
Par une requête et un mémoire complémentaire enregistrés le 14 octobre et le 6 novembre 2019 à 11h49, la société [redacted] *représentée par ses dirigeants et par Me* [redacted] *demande au juge des référés :*
1°) de suspendre les effets de l'arrêté du maire de Tauxigny-Saint-Bauld signé le 28 mai 2019 portant opposition à sa déclaration préalable en vue de l'installation d'une station de radiotéléphonie sur un terrain sis lieudit [redacted] *et ceux du rejet de son recours gracieux*

2. The municipality is not directly mentioned in the statement of facts nor the court's reasoning as it is not a party to the proceedings. Hence, the extraction returns a mismatch. In such cases, however, the municipality concerned is mentioned in the outcome section. By extracting the municipality several times, the number of observations with precise location is maximized.
   - **Text provided for evaluation by LLM:**
Vu la requête, enregistrée le 26 février 2008, présentée par M. [redacted]*, élisant domicile* [redacted] *à Saint Sauveur (70300) ;*
M. [redacted] *conteste un avis défavorable émis par la direction départementale des services vétérinaires du département de la Haute-Saône sur sa demande de certificat d'urbanisme ;*
Vu le code de justice administrative ;
Considérant qu'aux termes de l'article R. 222-1 du code de justice administrative : ...les présidents de tribunal administratif... peuvent, par ordonnance ...
4°) Rejeter les requêtes manifestement irrecevables, lorsque la juridiction n'est pas tenue d'inviter leur auteur à les régulariser ou qu'elles n'ont pas été régularisées à l'expiration du délai imparti par une demande en ce sens ... » ;
qu'aux termes de l'article R. 421-1 du même code : « Sauf en matière de travaux publics, la juridiction ne peut être saisie que par voie de recours contre une décision... »
Considérant que la requête susvisée est dirigée contre un avis défavorable de la direction départementale des services vétérinaires du département de la Haute-Saône relatif à une demande de certificat d'urbanisme ; que cet avis, qui n'a pas le caractère d'une décision administrative faisant grief, est insusceptible de recours devant la juridiction administrative ; qu'en conséquence, la requête de M. [redacted] *n'est pas recevable et doit être rejetée en application des dispositions susvisées de l'article R. 222-1 du code de justice administrative ;*

3. The date extracted is the date at which the judgment was read (4 July 2000), which is stated in the header of the decision, rather than the date of the hearing (20 June 2000). The hearing date only appears at the very end of the decision, in the sentence mentioning the deliberation.
   - **Text provided for evaluation by LLM:**
Jugement lu le 4 JUILLET 2000 ————— M. [redacted] *————— N°* [redacted] *RÉPUBLIQUE FRANÇAISE ————— TRIBUNAL ADMINISTRATIF DE CAEN (CALVADOS-MANCHE-ORNE) ————— AU NOM DU PEUPLE FRANÇAIS Le Tribunal Administratif de Caen 2ème Chambre*
Vu, enregistrée au greffe le 22 octobre 1999 sous le n° [redacted] *la requête présentée pour M.* [redacted] *demeurant* [redacted] *; M.* [redacted] *demande au Tribunal d'annuler l'arrêté en date du 19 février 1999 par lequel le préfet du Calvados a délivré un permis de construire un bâtiment agricole à M.* [redacted] *[···]*

   - **Actual information (end of the decision):**
Article 1er : L'arrêté en date du 19 février 1999 du préfet du Calvados est annulé.
Article 2 : L'Etat paiera à M. [redacted] *une somme de 4 000 F au titre des frais exposés par lui et non compris dans les dépens.*
Article 3 : Expédition du présent jugement sera notifiée à M. [redacted]*, au ministre de l'équipement, des transports et du logement et à M.* [redacted]*.*
Délibéré, en la même formation qu'à l'audience, le 20 juin 2000, étant présents : Mme [redacted]*, Président, Mme* [redacted] *et Mme* [redacted]*, Conseillers.*

4. The amount is not captured, and the variable is therefore set to 0. Article L761-1 of the Code of Administrative Justice, which orders the payment of irreducible costs, is mentioned, but the amount is written "1 000 (mille) euros", i.e. followed by its spelled-out value in brackets, a formulation that the extraction does not handle.
   - **Text provided for evaluation by LLM:**
Article 1er : Il n'y a pas lieu de statuer sur les conclusions à fin d'annulation de la requête de l'association [redacted]*.*
Article 2 : La commune de Montreuil versera à l'association [redacted] *une somme de 1 000 (mille) euros en application de l'article L. 761-1 du code de justice administrative.*
Article 3 : La présente ordonnance sera notifiée à l'association [redacted]*, à l'établissement public foncier d'Ile-de-France et à la commune de Montreuil.*

5. The case concerns a dispute over the creation of a micro-nursery, which is therefore not a housing unit. However, the text mentions that the nursery was created by renovating a housing unit, without specifying what happened to it. As this fact is mentioned in section 7 of the court's reasoning, the information is coded as 'TRUE' by the LLM.

- **Text provided for evaluation by LLM:**[32]

Vu la procédure suivante :
Par une requête et des mémoires, enregistrés les 15 mars et 18 décembre 2017 et 9 octobre 2018, la commune de Languidic, représentée par le [redacted], demande au juge des référés, dans le dernier état de ses écritures :
1°) de condamner conjointement et solidairement ou, à défaut, chacun pour son fait ou sa faute M. [redacted], la société [redacted], la société [redacted], la société [redacted] et la société [redacted], sur le fondement des dispositions de l'article R. 541-1
Considérant que la commune de Languidic a entrepris en 2010 une opération d'extension et de rénovation d'un bâtiment communal pour la création d'une micro-crèche ; que la maîtrise d'œuvre de ce projet a été confiée le 15 octobre 2010 à un groupement conjoint composé de M. [redacted], architecte, et de M. [redacted], économiste de la construction ; que le contrôle technique de cette opération était assurée par la société [redacted] en vertu d'une convention conclue le 26 novembre 2010 et portant notamment sur les missions LE relative à la solidité des existants et SEI relative à la sécurité des personnes dans les ERP ; que la réalisation des lots n° 1 Gros-œuvre, n° 2 Couverture étanchéité et n° 3 Menuiserie aluminium-Serrurerie a été attribuée respectivement aux sociétés [redacted], [redacted] et [redacted] par des actes d'engagement du 5 juillet 2011 ; que le 5 avril 2012, le maître de l'ouvrage a prononcé la réception sans réserve des travaux des lots n° 1 Gros-œuvre et n° 3 Menuiseries extérieures ; que le 28 octobre 2014, après avoir observé le fléchissement anormal des planchers de l'une des chambres situées à l'étage du bâtiment et effectué le décollement du revêtement de sol et de lattes de bois, le personnel de la commune constatera la présence de champignons de type mérule ; que l'expert, désigné par le juge des référés du tribunal administratif de Rennes à la demande de la commune, confirmera le 5 décembre 2014 la présence de mérule sous la forme de champignons lignivores s'attaquant aux lames de parquet de la chambre concernée et entrainant leur pourriture, ces désordres l'amenant à penser que l'ensemble des planchers du bâtiment était infesté de mérule et à recommander l'évacuation du bâtiment au regard des risques pour la sécurité des personnes et leur santé ; que lors des opérations d'expertise menées en 2015, les sondages destructifs révéleront une infestation partielle de mérule dans le plancher bas du rez-de-chaussée et une infestation très avancée dans le plancher bas du premier étage, les poutres étant affaiblies dans leur ensemble et l'une d'elle ayant complètement perdu sa résistance mécanique ; que, se fondant, à titre principal, sur l'engagement de la responsabilité décennale des constructeurs, et, à titre subsidiaire, sur la responsabilité contractuelle de l'architecte et de la société [redacted], la commune de Languidic demande au juge des référés de condamner conjointement et solidairement, ou, à défaut, chacun pour son fait ou sa faute, M. [redacted] et les sociétés [redacted], [redacted], [redacted] et [redacted] à lui verser, sur le fondement des dispositions de l'article R. 541-1 du code de justice administrative, une provision de 269 343,89 euros au titre du coût des travaux de reprises des désordres affectant l'ouvrage, une provision de 51 928,93 euros au titre des préjudices consécutifs à ces désordres et une provision de 16 348,94 euros au titre des frais d'expertise et de constat judiciaires ;
Sur l'existence d'une obligation non sérieusement contestable :
2. Considérant qu'aux termes de l'article R. 541-1 du code de justice administrative : « Le juge des référés peut, même en l'absence d'une demande au fond, accorder une provision au créancier qui l'a saisi lorsque l'obligation n'est pas sérieusement contestable (. . .) » ;
3. Considérant qu'il résulte des principes régissant la responsabilité des constructeurs que les désordres apparus dans le délai d'épreuve de dix ans, de nature à compromettre la solidité de l'ouvrage ou à le rendre impropre à sa destination dans un délai prévisible engagent leur responsabilité, même s'ils ne sont pas révélés dans toute leur étendue avant l'expiration du délai de dix ans ; que la responsabilité décennale d'un constructeur peut être recherchée à raison des dommages qui résultent de travaux de réfection réalisés sur les éléments constitutifs d'un ouvrage, dès lors que ces dommages sont de nature à compromettre la solidité de l'ouvrage ou à le rendre impropre à sa destination ; qu'ainsi, contrairement à ce que fait valoir la société [redacted], la commune de Languidic est parfaitement fondée à rechercher l'éventuelle responsabilité décennale des constructeurs ;
4. Considérant qu'en revanche, si la commune de Languidic produit les décisions prononçant la réception sans réserves des travaux des lots n° 1 Gros-Œuvre et n°3 Menuiseries extérieures attribués respectivement aux sociétés [redacted] et [redacted], il ne résulte pas de l'instruction que la réception des travaux du lot n° 2 Couverture étanchéité attribué à la société [redacted] ait été expressément prononcée ; qu'ainsi, seule la responsabilité contractuelle de la société [redacted] peut, en l'état de l'instruction, être recherchée par la commune de Languidic ;
5. Considérant que les sociétés [redacted] et [redacted] font valoir que la réception de leurs travaux intervenu sans réserve purge les vices et non-conformités apparents et que ces non-façons ne pouvaient qu'être visibles lors de la réception ; que, toutefois, en admettant que ces vices et non-conformités aient été apparents au moment de la réception le 5 avril 2012, leurs conséquences ne s'étaient pas encore révélées, la présence de la mérule étant seulement constatée à partir du 28 octobre 2014 ; qu'ainsi, la commune de Languidic reste fondée à rechercher la responsabilité décennale des sociétés [redacted] et [redacted] ;
6. Considérant qu'il résulte de l'instruction, notamment des rapports de l'expert judiciaire, que la micro-crèche municipale de la commune de Languidic présente une infestation partielle de mérule dans le plancher bas du rez-de-chaussée et une infestation très avancé dans le plancher bas du premier étage ; que la mérule s'est attaqué

32 The text 'statement of facts + court's reasoning' is over 3,000 words long; it has been truncated in the example to select only the parts of the text relevant to illustrating the error.

au parquet d'une des chambres situées à cet étage, les lames de cet parquet étant complètement pourries ; que ces champignons lignivores ont proliféré entre les solives du parquet ; que le démontage des doublages a mis en évidence l'infestation généralisée par de la mérule dont les ramifications se répandent sur l'ensemble des murs de la façade Ouest et sur l'ensemble du plancher bas de l'étage ; que les poutres de ce plancher sont affaiblies sur les encastrements en tête et que certaines d'entre elles ont totalement perdu leur résistance mécanique ; que ces désordres sont ainsi de nature à affecter la solidité de l'ouvrage et à le rendre impropre à sa destination ; que, dans ces conditions, la responsabilité des constructeurs est susceptible d'être engagée sur le fondement de la garantie décennale ;

7. Considérant qu'il ressort du rapport d'expertise que les désordres affectant cette micro-crèche sont la conséquence, en premier lieu, d'une absence de diagnostic préalable des existants alors que la mission de maîtrise d'œuvre comportait bien un élément relatif au diagnostic (DIAG) et que l'architecte reconnaît dans son mémoire en défense que les photos de l'ancien logement à rénover présentaient des tâches d'humidité apparentes au niveau de la fenêtre de la lucarne des combles, de l'aplomb de la souche de la cheminée Ouest des combles et au niveau des murs en soubassement de la cave ; que ce diagnostic préalable aurait permis à l'architecte de prévoir la démolition de l'ensemble des doublages existants ; que ces désordres résultent, en second lieu, d'un défaut de conception de l'architecte aboutissant à la création d'espaces confinés et non ventilés par la pose de doubles cloisons tout en conservant les anciens doublages et revêtements et les plinthes précédentes ainsi que de la pose d'un revêtement imperméable à l'air sur les parquets d'origine ; que ces désordres sont également la conséquence d'un manquement de l'architecte dans sa mission de suivi de l'exécution des travaux par les entreprises ; qu'il ressort, en effet, du rapport d'expertise que la société [redacted] n'a pas réalisé les bandes de dressement et les rejingots au niveau de la capucine des combles, prestations prévues aux articles 1.3.5.1 et 1.3.5.2 de son CCTP, et que le glacis sur l'appui de la capucine est mal réalisé ; que la baie de cette capucine présente également un défaut de pose étant simplement calfeutrée au silicone laissant s'infiltrer l'eau de pluie le long du mur intérieur alors que la société [redacted] devait les calfeutrements et les joints d'étanchéité Compriband ; que ces non-façons et malfaçons par les sociétés [redacted] et [redacted] sont à l'origine d'infiltrations favorisant la prolifération de la mérule ; que ces désordres sont aussi imputables à la société [redacted], contrôleur technique chargé notamment d'une mission LE relative à la solidité des existants, qui ne justifie pas avoir mis en garde le maître de l'ouvrage contre l'absence de diagnostic des existants et sur le défaut de conception des doubles cloisons ainsi que sur les non-façons et malfaçons réalisées par les entreprises, alors que ces vices ont fini par compromettre la solidité des parties anciennes de l'ouvrage, telles que les planchers ; que ces désordres sont, par conséquent, de nature à engager la responsabilité solidaire de M. [redacted], de la société [redacted], titulaire du lot n° 1 Gros-œuvre, de la société Réalu, titulaire du lot n° 3 Menuiseries extérieures, et de la société [redacted], contrôleur technique ;

High accuracy scores and example analysis hence confirm the database's reliability. The LLM demonstrates conservative behavior when information is unavailable in the text, leading to minimal hallucinations during extraction.

D.4 Sensitivity to number of examples provided

The number of examples provided in the prompt significantly impacts both extraction quality and cost. As illustrated in Figure D.1, which shows accuracy for variables evaluated on statement of facts, using only 2 or 5 examples fails to achieve optimal performance levels. However, between 10 and 15 examples, the added value becomes unclear, as excessively long prompts may lead to hallucinations while increasing costs by 50%. It seems that the performance gains plateau beyond 10 examples, suggesting diminishing returns. Therefore, we selected 10 examples as the optimal balance between performance quality and computational efficiency, providing sufficient context for accurate extraction without introducing unnecessary complexity or cost overhead.

D.5 Measure of model stochasticity

Model temperature is fixed at 0 to ensure deterministic outputs. We evaluate reproducibility by testing whether identical prompts consistently produce the same results across 50 iterations, focusing on variables extracted from statement of facts. Figure D.2 demonstrates that most variables exhibit very low variability: the interquartile range is approximately 0.6%. For 3 out of 6 tested variables, this range is even smaller at less than 0.1%. These results confirm the stability and reliability of the extraction process, indicating that the model produces consistent outputs when given identical inputs under controlled conditions (Törnberg, 2024).

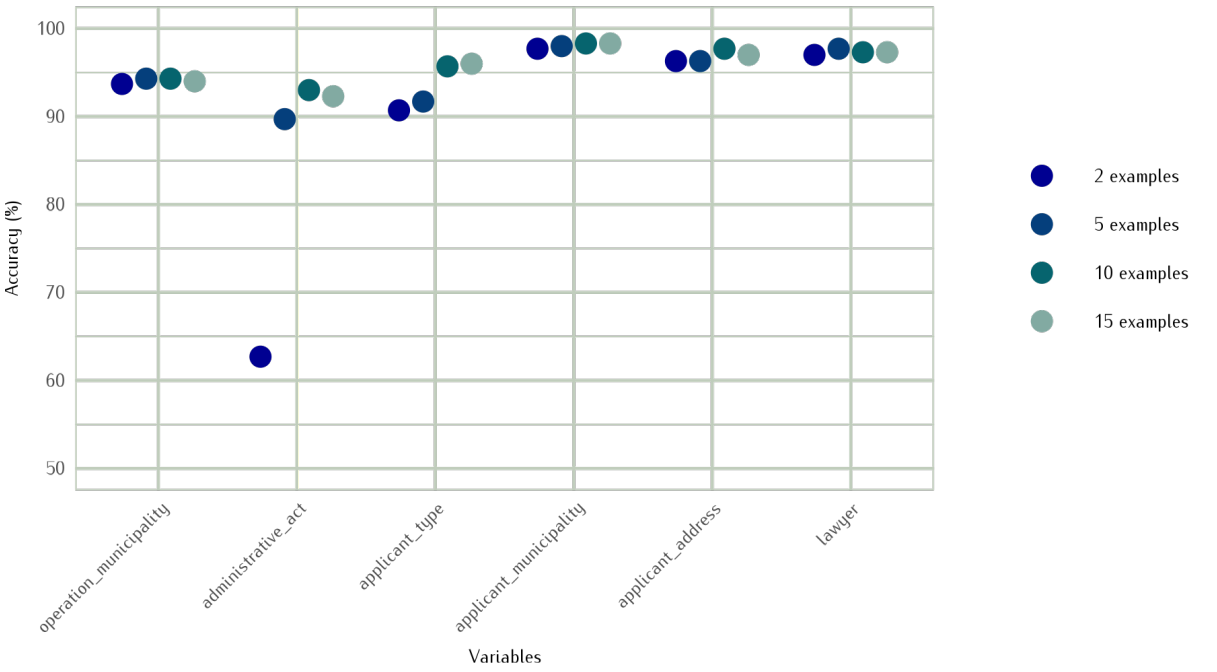


Figure D.1: Sensitivity analysis on number of examles

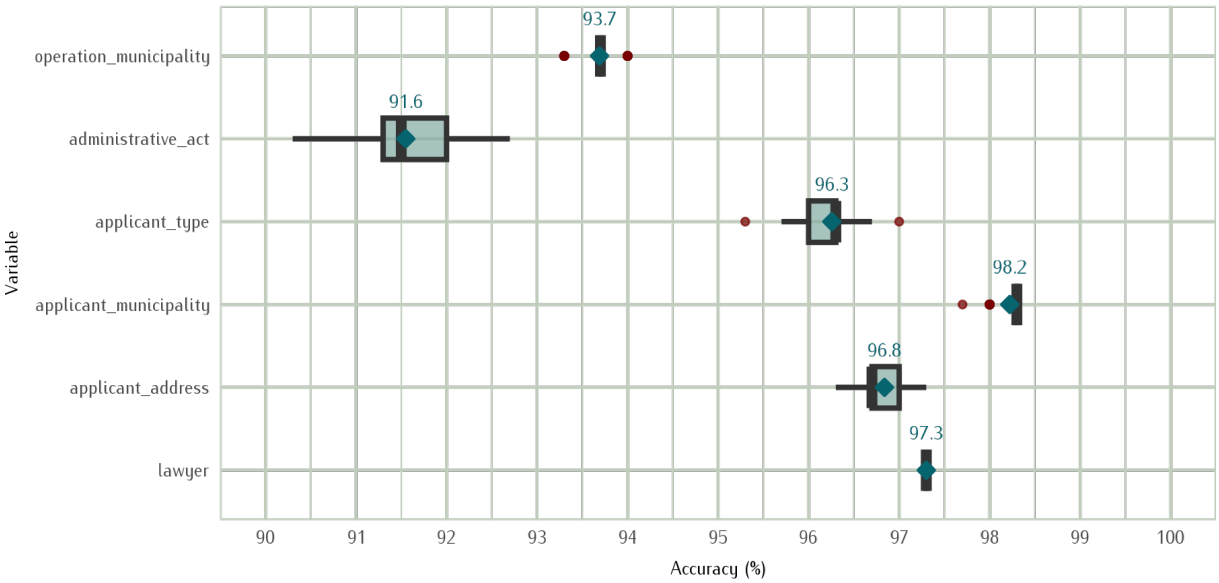


Figure D.2: Distribution of accuracy over 50 runs

Notes: Green diamond: mean. Box plot: Q1, median, Q3. Red dots: outliers. Data based on 50 runs per variable. The score for operation municipality is slightly lower than the final performance presented in Table D.1, which corresponds to the extraction in 'statement of facts + court's reasoning'.

D.6 Added value of text partitioning

We tested the effectiveness of segmenting documents into targeted sections rather than processing complete texts, as the latter approach can lead to hallucinations and reduced extraction accuracy when dealing with lengthy documents. Our sectioned approach demonstrates superior performance across most variables compared to full-text processing (see Table D.4). Notably, several variables show substantial improvements: `operation_address` accuracy increases from 70.7% to 82.7%, `administrative_act` improves from 87.7% to 93.0%, and `construction_type` rises from 75.7% to 83.3%. The sectioned approach also shows better performance for variables like `lawyer` (91.0% to 97.3%) and `social_housing_number` (85.7% to 100.0%). While a few variables like `date_application` (extracted with regex to reduce cost) and `plot_classification` perform slightly better with full-text processing, the overall pattern clearly favors the segmented approach. Most importantly, error metrics for numerical variables are significantly lower with segmentation, as evidenced by the dramatic reduction in `compensation` RMSE from 13,614.70 to 130.34, demonstrating that targeted section processing effectively reduces hallucinations and improves extraction reliability.

Variable	Accuracy	MAE	RMSE	Precision	Recall	F1 micro	F1 macro
`date_application`	99.7%	0.00	0.00	-	-	-	-
`date_hearing`	94.0%	2.85	27.63	-	-	-	-
`operation_municipality`	98.3%	-	-	-	-	-	-
`operation_address`	70.7%	-	-	-	-	-	-
`administrative_act`	87.7%	-	-	90.9%	89.7%	90.3%	79.6%
`initial_outcome`	89.3%	-	-	89.3%	89.3%	89.3%	88.9%
`plot_classification`	98.0%	-	-	98.0%	98.0%	98.0%	78.0%
`applicant_type`	96.3%	-	-	97.4%	98.3%	97.9%	95.8%
`applicant_municipality`	97.0%	-	-	-	-	-	-
`applicant_address`	95.0%	-	-	-	-	-	-
`lawyer`	91.0%	-	-	91.0%	91.0%	91.0%	86.7%
`housing`	93.3%	-	-	93.3%	93.3%	93.3%	93.4%
`housing_number`	82.1%	0.81	4.00	-	-	-	-
`social_housing`	98.0%	-	-	98.0%	98.0%	98.0%	78.0%
`social_housing_number`	85.7%	3.71	9.83	-	-	-	-
`construction_type`	75.7%	-	-	77.7%	81.6%	79.6%	75.7%
`outcome`	95.7%	-	-	98.0%	97.2%	97.6%	90.0%
`compensation`	97.7%	842.31	13,614.70	-	-	-	-
`irreducible_costs`	96.3%	72.13	470.41	-	-	-	-

Table D.4: Assessment of performance when using one prompt on the full text of files

E DERIVATIONS OF THE LITIGATION MODEL

This appendix derives the settlement condition used in section 4, shows how access to a lawyer and the opportunity cost of time shape the decision to file, and writes out the filing decision of an association.

E.1 The settlement condition

We adapt the game of Rosenberg and Shavell (1985). The neighbor files at cost c_i and demands a settlement s_{ij} in exchange for her withdrawal. The developer either pays or lets the case run, and the neighbor then pursues the case or withdraws. Both parties are risk-neutral, know all the variables and make no threat they would not carry out, so we solve the game backward.

Once the case runs, pursuing it costs the neighbor nothing more, since c_i is paid at filing, and yields $\pi_{ij}B_{ij}$. She therefore pursues whenever she suffers from the project ($B_{ij} \geq 0$), and her threat to do so is credible. Letting the case run then costs the developer M_j, so it accepts any demand $s_{ij} \leq M_j$ and rejects any higher one. The neighbor demands the most the developer accepts, $s_{ij} = M_j$, unless she prefers the case to run, which yields her $\pi_{ij}B_{ij}$. The parties therefore settle if and only if

$$M_j > \pi_{ij}B_{ij}, \qquad s_{ij} = M_j.$$

Her value of filing is $G_{ij} = \max(\pi_{ij}B_{ij},\, M_j)$, and she files if and only if $G_{ij} \geq c_i$.

E.2 Lawyer access and the cost of litigation

We decompose the litigation cost c_i to show how access to a lawyer and the opportunity cost of time shape the filing condition $\max(\pi_{ij}B_{ij},\, M_j) \geq c_i$ of section 4. Litigating takes time t, valued at the neighbor's wage w_i. Without a lawyer, the permit is annulled with probability q_N and $c_N = w_i t$. With a lawyer, it is annulled with probability $q_L > q_N$ and $c_L = f + w_i t$, where f is the fee. We observe empirically that having a lawyer doubles the probability of winning when controlling for case characteristics, court and year of hearing (Figure A.2). Let π_k and M_k denote π_{ij} and M_j evaluated at q_k, for $k \in \{N, L\}$. A lawyer raises both what a filing secures and what a pending case costs the developer:

$$\pi_L - \pi_N = (q_L - q_N)\big(1 - g(T)\big), \qquad M_L - M_N = (q_L - q_N)\big(1 - \gamma(T)\big)\,\Pi_j.$$

Both gains shrink as delays lengthen, because a growing share of the value of a case then comes from delay, whatever its merits. As the time cost is incurred either way, the neighbor hires a lawyer when the relevant gain exceeds the fee. We consider cases whose outcome, judgment or settlement, does not depend on representation.

Cases that go to judgment. If $\pi_k B_{ij} \geq M_k$, the neighbor files when $B_{ij} \geq w_i t/\pi_N$ without a lawyer and $B_{ij} \geq (f + w_i t)/\pi_L$ with one. She hires a lawyer if

$$(q_L - q_N)\big(1 - g(T)\big)\, B_{ij} > f,$$

so counsel pays off only for large stakes. Lawyer access brings in additional filers only if it lowers the filing threshold, that is if

$$\frac{\pi_L}{\pi_N} > \frac{f + w_i t}{w_i t}.$$

This condition fails when delays are long, as π_L/π_N then tends to 1. It is easier to meet at high wages: the marginal filer then has a larger stake, and the gain from counsel scales with the stake.

Settled cases. If $M_k > \pi_k B_{ij}$, the neighbor files when the settlement covers her cost, whatever her stake: $M_N \geq w_i t$ without a lawyer, $M_L \geq f + w_i t$ with one. She hires a lawyer if

$$(q_L - q_N)\big(1 - \gamma(T)\big)\,\Pi_j > f.$$

The gain from counsel now scales with the developer's stake, so a lawyer pays off against large projects, even for a neighbor with little at stake.

Income and wealth. Filing thresholds rise with the wage in both cases, so among neighbors who can afford counsel, those with a low opportunity cost of time, such as retirees, have the strongest incentives to file. Suppose now that, of two neighbors with wages $\overline{w} > \underline{w}$, only the high-income one can afford a lawyer. She has the stronger incentive to file if

$$\frac{\pi_L}{\pi_N} > \frac{f + \overline{w}t}{\underline{w}t}$$

in a case that goes to judgment, and if

$$(q_L - q_N)\big(1 - \gamma(T)\big)\,\Pi_j > f + (\overline{w} - \underline{w})\,t$$

in a settled case. The first condition does not depend on the stakes and requires counsel to multiply what a filing secures by more than the ratio of the two costs; the second holds for large enough projects.

E.3 Associations

An association A can challenge permit j only if its statutes were filed at least one year before the permit application was posted (Article L600-1-1 of the Urban Planning Code). It pools the stakes of the neighbors $i \in A$ it represents, and bears a cost of litigation c_A and a cost of organizing F_A, such as gathering members and deciding to act. Since it cannot be paid in cash to withdraw (Article L600-8 of the Urban Planning Code), the developer's cost of a pending case M_j plays no role in its decision, and its cases go to judgment. Association A files if and only if

$$\sum_{i \in A} \pi_{ij} B_{ij} \geq c_A + F_A.$$

Because $\pi_{ij} = \pi_j$ is the same for all neighbors of project j, with n_A members with the same stake B_j this condition becomes

$$n_A \geq \frac{c_A + F_A}{\pi_j B_j}.$$

An association can therefore bring a case that no single member would file ($\pi_j B_j < c_i$ for every member) once its members are numerous enough, which makes association litigation more likely where neighbors are many. For environmental associations, whose standing derives from their statutory purpose rather than from their members' property, B_{ij} is best read as their members' valuation of the natural amenity that the project removes. Associations may still reach an agreement through a modification of the project, which we do not model.